



Manufacturing Process

Complete Zero-to-Hero Exam Notes — All Units + PYQs with Answers + Sample Questions

- Casting
- Welding
- Metal Forming
- Machining
- CNC
- EDM/ECM
- USM/AJM
- LBM/EBM
- Additive Mfg

Compiled by **Akshit Raj** · [mujstella.in](#) · For more PYQs & resources, visit the website or download the MUJstella app on PlayStore



TABLE OF CONTENTS

- U1

Casting: Pattern & Allowances
- U2

Moulding Sand & Properties
- U3

Mould Making, Core & Gating System
- U4

Casting Methods
- U5

Welding: Concepts & Classification
- U6

Gas Welding & Flames
- U7

Electric Arc & Submerged Arc Welding
- U8

Resistance Welding
- U9

Soldering & Brazing
- U10

Metal Forming: Hot & Cold Working
- U11

Forging, Rolling, Extrusion, Drawing
- U12

Machining: Principles & Chips
- U13

Cutting Tools & Tool Geometry
- U14

Lathe: Parts & Operations
- U15

Milling & Drilling
- U16

CNC: Programming & G/M Codes
- U17

USM & AJM
- U18

ECM & EDM
- U19

LBM & EBM
- U20

Additive Manufacturing
- ★

Formula & Data Sheet
- ★

CO-wise Exam Guide

⚡ FORMULA & KEY DATA SHEET — QUICK REFERENCE

TOOL LIFE — TAYLOR'S EQUATION

$$V \times T^n = C$$

V = cutting speed (m/min) · T = tool life (min)
· n = Taylor's exponent (0.1–0.5) · C = Taylor's constant (speed for 1 min life)

RESISTANCE WELDING — HEAT GENERATED

$$H = I^2 \times R \times t$$

H = heat (joules) · I = current (amperes) · R = resistance (ohms) · t = time (seconds)

CUTTING SPEED

$$V = \pi \times D \times N / 1000$$

V = cutting speed (m/min) · D = workpiece diameter (mm) · N = spindle speed (RPM)

MATERIAL REMOVAL RATE (TURNING)

$$MRR = V \times f \times d$$

V = cutting speed · f = feed (mm/rev) · d = depth of cut (mm) → result in mm³/min

SHRINKAGE ALLOWANCE — KEY VALUES

Cast Iron ≈ 1% (10 mm/m)

Aluminium ≈ 1.6% · Steel ≈ 2% · Brass ≈ 1.5%
· Bronze ≈ 1.0–1.6%

ECM — FARADAY'S LAW (MRR)

$$MRR = (I \times A) / (F \times \rho \times z)$$

I = current · A = atomic weight · F = Faraday's constant (96,485 C/mol) · ρ = density · z = valence

AC WELDING — OCV RANGE

80 – 110 V (open circuit)

DC welding: 8–25 V · SAW current: 200–2000 A · ECM voltage: 8–25 V · EDM voltage: 50–300 V

BRAZING VS SOLDERING TEMPERATURE

Brazing > 450°C · Solder < 450°C

Soft solder: 37% Pb + 63% Sn, melts at 183°C · Hard solder: Cu+Sn+Ag, 600–900°C

CNC — KEY G-CODES TO MEMORISE

G00 G01 G02 G03 G90 G91

G00=Rapid · G01=Linear feed · G02=CW arc · G03=CCW arc · G90=Absolute · G91=Incremental

CNC — KEY M-CODES TO MEMORISE

M03 M05 M06 M08 M30

M03=Spindle ON CW · M05=Spindle OFF · M06=Tool Change · M08=Coolant ON · M30=End+Rewind

USM FREQUENCY

20 – 40 kHz vibration

Abrasive grain size: 10–50 μm · AJM gas pressure: 2–8 kg/cm² · EBM voltage: 25–200 kV · EBM vacuum: 10⁻⁴ torr

TOOL SIGNATURE FORMAT (7 PARAMETERS)

α-γ-θe-θs-Ce-Cs-R

Back rake – Side rake – End relief – Side relief – End cutting edge – Side cutting edge – Nose radius
Example: 0-7-6-8-15-16-0.8

ROLLING — PRODUCT CLASSIFICATION

CENTRIFUGAL CASTING — SPEED

GAS WELDING FLAME RATIOS (O₂:C₂H₂)

Bloom ≥150×150 mm (sq) Slab: rect, width ≥250mm, thickness ≥40mm · Billet: sq, sides ≥40mm	Horizontal: 200–1000 RPM Vertical: moderate RPM · Core baking temp: 200–250°C · Investment casting preheat: ~1000°C	Neutral 1:1 · Oxidising 1.15–1.5:1 Carburising = excess acetylene (no fixed ratio) · Flame temp ≈ 3300°C · OAW = Oxy-Acetylene Welding
DIE CASTING — TEMPERATURE CUTOFF Hot Chamber: <650°C metals Hot: Lead, Tin, Zinc · Cold: Al, Cu, Brass (>700°C) · EDM spark temp: 8000–12000°C locally		



CO-WISE EXAM QUESTION GUIDE + PREDICTION RATINGS

Questions are tagged **HOT** (appeared multiple times / very likely), **WARM** (appeared once / likely), or **POSSIBLE** (in syllabus / expected). All exam sections mapped by CO.

CO1 — CASTING & MOULD MAKING

Units 1–4 · Topics: Pattern allowances, Sand properties, Gating system, Casting methods

- Pattern allowances
- Types of sand
- Sand properties
- Gating system
- Riser purpose
- Core making
- Centrifugal casting
- Investment casting
- Die casting

- HOT** State the 5 pattern allowances in casting. (4 marks) — appeared Re-sess Nov 2025
- HOT** Describe centrifugal casting with diagram. (4 marks) — appeared MTE Sep 2025, Re-sess
- HOT** Prepare diagram of gating system + explain components. (6 marks) — ETE Nov 2025
- HOT** Properties of moulding sand (any 4). (4 marks) — MTE Sep 2025
- WARM** What is the role of core in casting? (4 marks) — MTE Feb 2025
- WARM** Investment casting process with neat sketches. (9 marks) — MTE Feb 2025
- WARM** Differentiate hot vs cold chamber die casting. (4 marks)
- POSSIBLE** Name 2 types of sand for mould making. (2 marks) — ETE Nov 2025
- POSSIBLE** What is rapping allowance? Why is it negative? (2 marks)

CO2 — WELDING PROCESSES

Units 5–9 · Topics: Welding classification, Gas welding, Arc welding, Resistance welding, Brazing

- Welding classification
- Gas flames
- SMAW / SAW
- Resistance welding types
- Spot / Seam welding
- Brazing vs soldering
- Shielding gas
- Electrodes

- HOT** Describe 4 methods of resistance welding. (8 marks) — MTE Sep 2025 + Feb 2025
- HOT** Describe submerged arc welding (SAW) process. (6 marks) — ETE Nov 2025
- HOT** Types of flames in gas welding with neat sketches. (4 marks) — Re-sess Apr 2026
- WARM** Differentiate brazing and soldering. (3 marks) — MTE Feb 2025
- WARM** Classify welding processes + explain any two. (4 marks) — MTE Feb 2025
- WARM** Welding vs brazing — one prime difference. (2 marks) — ETE Nov 2025
- WARM** Function of shielding gas in welding. (2 marks) — Re-sess Nov 2025
- POSSIBLE** Straight vs reversed polarity in DC arc welding. (4 marks)

CO3 — METAL FORMING & MACHINING

Units 10–15 · Topics: Hot/cold working, Forging/Rolling/Extrusion, Machining, Lathe, Milling/Drilling

- Hot vs Cold working
- Blanking vs Punching
- Rolling mills
- Extrusion types
- Wire vs deep drawing
- Chip types
- Tool angles
- Lathe operations
- Milling operations
- Drilling ops

- HOT** Differentiate hot working vs cold working (table). (4 marks) — MTE Sep 2025
- HOT** Blanking vs Punching in sheet metal. (4 marks) — MTE Sep 2025
- HOT** Discuss 4 lathe operations + tools used. (6 marks) — ETE Nov 2025
- HOT** Metal rolling process, features, applications. (10 marks) — ETE Nov 2025
- WARM** Differentiate wire drawing vs extrusion. (4 marks) — Re-sess Nov 2025

-  **WARM** Drawing vs deep drawing. (4 marks) — Re-sess Nov 2025
-  **WARM** Operations of milling machine. (4 marks) — Re-sess Apr 2026
-  **WARM** Parts of drilling machine. (4 marks) — Re-sess Nov 2025
-  **POSSIBLE** List 2 angles of single-point cutting tool. (2 marks) — ETE Nov 2025
-  **POSSIBLE** What is plastic deformation? (2 marks) — MTE Sep 2025

CO4 — CNC PROGRAMMING

Unit 16 · Topics: CNC components, G-codes, M-codes, Part programming

- CNC components
- G00/G01/G02/G03
- G90/G91
- M03/M05/M06/M30
- Part program writing
- Slot machining

-  **HOT** CNC components + write part program for 1.5mm slot in 60×30×6mm. (10 marks) — ETE Nov 2025
-  **HOT** Sketch of CNC machine components + describe. (9 marks) — Re-sess Apr 2026
-  **WARM** Applications of CNC machines. (4 marks) — Re-sess Nov 2025
-  **WARM** List 2 miscellaneous (M-code) functions. (2 marks) — Re-sess Apr 2026
-  **POSSIBLE** G01 and G03 stand for? (2 marks) — ETE Nov 2025
-  **POSSIBLE** Difference between G90 and G91. (2 marks)

CO5 — ADVANCED MACHINING & ADDITIVE MANUFACTURING

Units 17–20 · Topics: USM, AJM, ECM, EDM, LBM, EBM, Additive Manufacturing (FDM, SLA, SLS)

- USM vs AJM
- ECM principle
- EDM process
- LBM vs EBM
- Advanced vs conventional
- FDM process
- Additive mfg applications
- Machining vs AM

-  **HOT** Describe EDM process, parameters, applications. (10 marks) — ETE Nov 2025
-  **HOT** Explain FDM process of additive manufacturing. (6 marks) — ETE Nov 2025
-  **HOT** ECM — how desired shape is obtained. (8 marks) — Re-sess Nov 2025
-  **HOT** Chart of unconventional machining: energy, features, applications. (6 marks) — ETE Nov 2025
-  **WARM** Applications of AM in 2 fields. (8 marks) — Re-sess Nov 2025
-  **WARM** USM vs AJM comparison. (2 marks) — ETE Nov 2025
-  **WARM** Machining vs Additive Manufacturing. (3 marks) — Re-sess Apr 2026
-  **WARM** Compare conventional vs non-conventional machining. (4 marks) — Re-sess Apr 2026
-  **POSSIBLE** LBM principle and process. (4 marks)
-  **POSSIBLE** Why does EBM require vacuum? (2 marks)

UNIT 1

Casting — Pattern & Allowances

| WHAT IS CASTING?

Casting is a manufacturing process in which molten metal is poured into a mold, allowed to solidify, and then removed to form a desired shape. It is one of the oldest and most versatile manufacturing processes.

 **Simple definition:** Melt metal → pour into mold → solidify → remove → finished part. The mold gives the part its shape.

Steps in Casting Process

- Mould is prepared (sand mould, die mould, etc.)
- Metal is heated above its melting point
- Liquid metal is poured into the mold
- Metal solidifies into the shape of the mold
- Cast part is removed from mold
- Post-casting operations: cutting, clean-up, machining

| PATTERN

A **pattern** is a replica of the casting to be made. Using the pattern, impressions (mould cavities) are formed in damp sand to create the mould. The pattern is slightly larger than the final casting to account for allowances.

Pattern Materials

MATERIAL	PROPERTIES	USE CASE
Wood	Easy to work, readily available, can be shaped	Small number of castings, prototype work
Metal	Does not change shape with moisture, durable	Mass production, high-accuracy castings
Plastics	Do not absorb moisture, smooth surface, easy withdrawal	Complex shapes, medium production
Plaster	High compressive strength	Intricate patterns

Types of Patterns

TYPE	DESCRIPTION	USE
Single Piece	Made from one piece, no loose parts, inexpensive, simplest type	Simple shapes, prototype
Split Pattern	Most widely used; pattern split into 2 parts — one in cope, one in drag; aligned by dowel pins	Intricate castings
Cope & Drag Pattern	Similar to split; cope and drag halves prepared separately with gating/riser systems	Large castings
Match Plate Pattern	Extension of cope & drag; both halves mounted on single plate on either side; includes gating system	Mass production
Loose-Piece Pattern	Used when contour prevents direct withdrawal; obstructing part held as loose piece	Complex undercuts
Skeleton Pattern	Made of wood strips; sand packed around it	Very large, one-off castings
Gated Pattern	Made of metal; sections between patterns serve as runners/gates; can make multiple castings at once	Mass production

| PATTERN ALLOWANCES

Pattern allowances are intentional dimensional differences between pattern and final casting to compensate for various factors during casting.

#	ALLOWANCE	PURPOSE	DETAILS
1	Shrinkage Allowance	Compensates for metal contraction during cooling	Pattern is made larger; amount varies with metal type (e.g., cast iron ~1%, aluminium ~1.6%)
2	Machining/Finishing Allowance	Extra metal added to surfaces that will be machined later	Given in addition to shrinkage allowance; depends on machining method
3	Draft / Taper Allowance	Facilitates easy withdrawal of pattern from mould	Slight taper given on vertical surfaces parallel to direction of withdrawal; on both internal & external surfaces
4	Rapping / Shake Allowance	Compensates for cavity enlargement during pattern removal	Negative allowance — when pattern is rapped (struck sideways) to loosen it, cavity slightly increases
5	Distortion / Camber Allowance	Compensates for unequal contraction in irregular castings	Opposite distortion deliberately built into pattern; important for U-shapes, T-shapes


Exam tip: The 5 allowances appear frequently in both 2-mark and 4-mark questions. Remember: Shrinkage, Machining, Draft, Rapping, and Distortion. Rapping is the **ONLY** negative allowance.


PREVIOUS YEAR QUESTIONS — UNIT 1 (CASTING: PATTERN & ALLOWANCES)

RE-SESSIONAL NOV 2025
4 marks · CO1

Q4. State different allowances used in casting.

The five pattern allowances in casting are:

- Shrinkage Allowance:** Pattern made larger to compensate for metal contraction during cooling from molten to room temperature. Amount varies — cast iron ≈1%, Al ≈1.6%.
- Machining/Finishing Allowance:** Extra metal added on surfaces to be machined. Given in addition to shrinkage allowance.
- Draft/Taper Allowance:** Slight taper on vertical walls parallel to direction of withdrawal to allow easy pattern removal from mould.
- Rapping Allowance (Negative):** When pattern is rapped sideways to loosen it, cavity enlarges slightly. A negative allowance compensates for this.

- **Distortion Allowance:** For irregular castings with non-uniform cooling, an opposite distortion is built into the pattern.

ETE NOV 2025 2 marks · CO1

QA1. Name any two types of sand used for mould making.

- **Green Sand:** Contains silica sand, clay, and water. Possesses good strength and permeability. Most commonly used.
- **Dry Sand:** Stronger and more thermally stable than green sand. Used for larger, heavier castings.
- (Also acceptable: Loam Sand, Facing Sand, Backing Sand, Core Sand)

MTE SEP 2025 2 marks · CO1

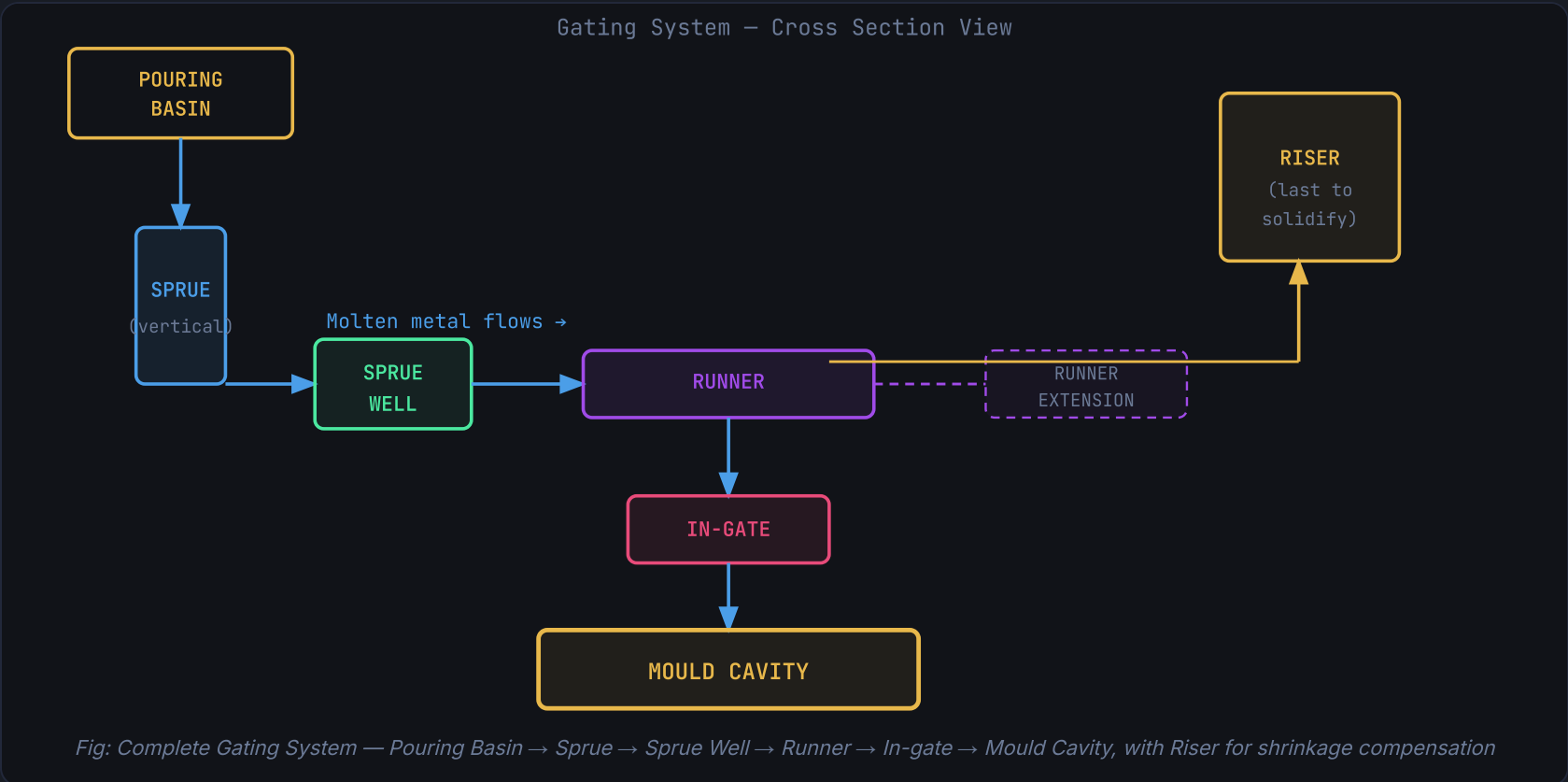
Q1. What is Casting?

Metal casting is a manufacturing process in which molten metal is poured into a mold, allowed to solidify, and then removed to form a desired shape.

ETE NOV 2025 6 marks · CO1

QB1. Prepare a diagram of gating system used in casting and explain function of its various components.

Gating System Components and Functions:



- **Pouring Basin:** Funnel-shaped reservoir at top where molten metal is initially poured from crucible. Reduces turbulence.
- **Sprue:** Vertical channel connecting pouring basin to runner. Controls flow rate of metal.
- **Sprue Well:** Small reservoir at bottom of sprue. Regulates flow into runner and reduces turbulence.
- **Runner:** Horizontal channel(s) distributing molten metal from sprue to gates.
- **Runner Extension:** Extended part of runner to feed multiple gates or collect impurities.
- **In-gate (Gate):** End opening through which molten metal enters the mold cavity.
- **Riser:** Metal reservoir holding extra molten metal to compensate for shrinkage during solidification. Must be last part to solidify.

Functions of Gating System: Controls flow rate, minimizes turbulence, prevents air entrapment, guides metal to complex areas, provides shrinkage compensation through riser.

RE-SESSIONAL NOV 2025 2 marks · CO1

Q1. What is the purpose of providing a riser in a casting mold?

A **riser** acts as a metal reservoir that holds extra molten metal. It ensures continuous flow of metal into the mold cavity as the casting material shrinks during solidification, thereby:

- Preventing shrinkage cavities and porosity voids
- Maintaining the desired shape and dimensions of the casting
- Should be the last part of the mold to solidify

MTE SEP 2025 4 marks · CO1

Q7. Describe centrifugal casting method.

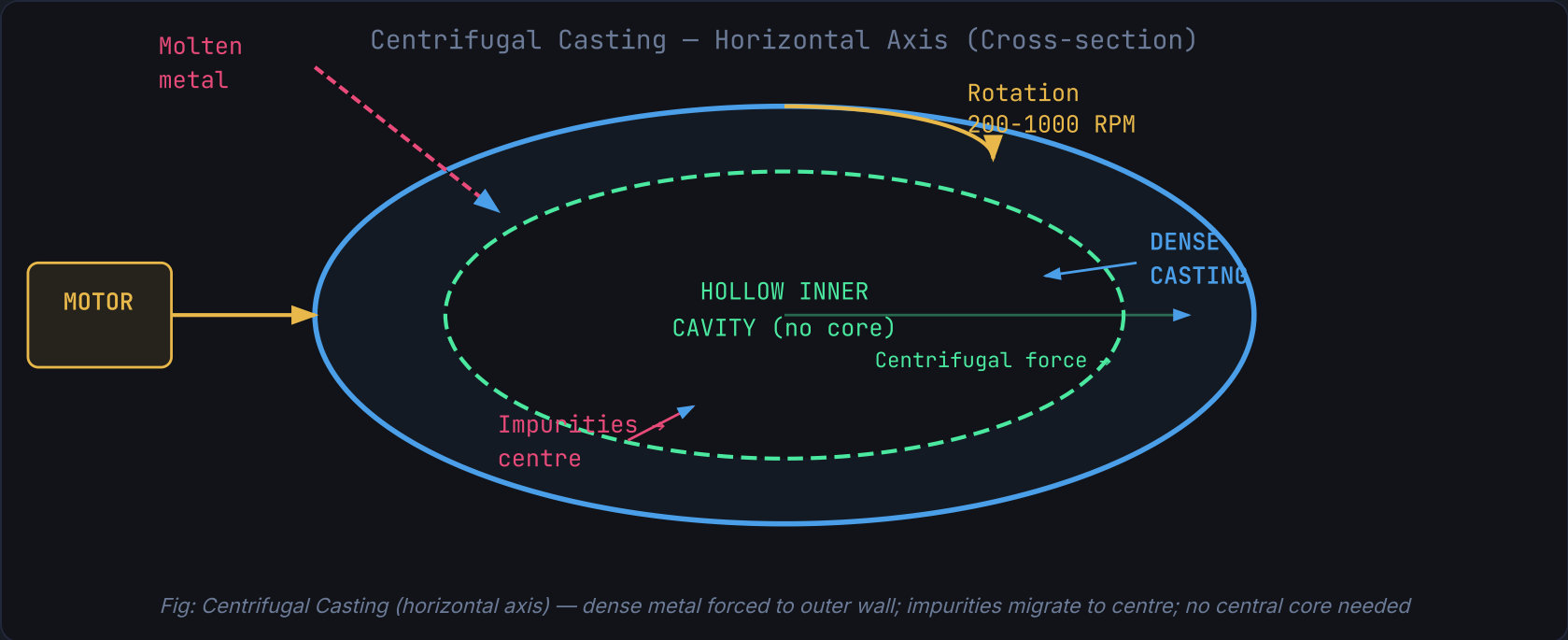
Centrifugal Casting is a metal casting process that utilizes centrifugal force to distribute molten metal into a rotating mold. **Process:**

- Molten metal is introduced into a rotating mold (sand, metal, or graphite)
- Mold is set up and rotated along vertical (moderate RPM) or horizontal axis (200–1000 RPM)

- Mold is coated with refractory coating
- Centrifugal force holds metal against mold wall until solidified
- During cooling, lower-density impurities rise towards center of rotation

Products: Pipes, tubes, cylindrical/symmetrical parts with rotational symmetry.

Advantages: Uniform metal distribution, reduced defects, dense outer surface, no central core needed for hollow cylinders.



RE-SESSIONAL APR 2026 9 marks · CO2

Q8. Prepare sketch of components of CNC machine and describe its various components. (Note: This appears in re-sessional — refer Unit 16 for CNC answer)

See Unit 16 — CNC section for full detailed answer.

💡 EXPECTED/SAMPLE QUESTIONS — UNIT 1

- ⚡ WARM

1. Differentiate between cope and drag pattern and match plate pattern. (4 marks)
- 🔥 HOT

2. What is the purpose of draft allowance in casting? (2 marks)
- ⚡ WARM

3. List the steps involved in green sand mould making. (4 marks)
- 💧 LIKELY

4. What is a split pattern? Why are dowel pins used? (2 marks)
- 🔥 HOT

5. Why is shrinkage allowance given in a pattern? Name any two metals and their shrinkage amounts. (2 marks)

UNIT 2

Moulding Sand & Its Properties

CLASSIFICATION OF MOULDING SAND

TYPE	COMPOSITION	PROPERTIES / USE
Green Sand	Silica sand + clay + water	Possesses good strength and permeability. Most widely used. "Green" = presence of moisture, not colour.
Dry Sand	Green sand dried in oven	Stronger and more thermally stable than green sand. Used for larger, heavier castings.
Loam Sand	High clay content (~50%)	Used for large castings with skeleton patterns.
Facing Sand	Fine sand with high refractoriness	Directly contacts the casting; gives smooth surface finish.
Backing Sand	Old, reconditioned sand	Supports facing sand; fills rest of mould. Contains permeability.
Core Sand	Sand + binders (linseed oil, resins)	Used to make cores. Must have high dry strength.
Parting Sand	Fine dry sand (no clay)	Sprinkled at parting line to prevent sand halves sticking together.
System Sand	High quality sand	Used in machine moulding; needs high strength, permeability, refractoriness.

KEY COMPONENTS OF GREEN SAND

- **Silica Sand (SiO₂):** Primary mold material — provides refractoriness. Typical range: 80–92%
- **Clay (Bentonite/Kaolinite):** Acts as binder — holds sand grains together. Typical range: 6–12%
- **Water:** Activates clay binder — enables it to hold grains. Typical range: 3–6%
- **Other additives:** Sea coal, silica flour — improve flowability, permeability, surface finish

PROPERTIES OF MOULDING SAND

PROPERTY	DESCRIPTION	WHY IMPORTANT
Porosity (Permeability)	Ability to allow gases/moisture to escape when molten metal is poured	Prevents gas defects (blow holes, porosity) in casting
Adhesiveness	Sand particles capable of adhering to sides of moulding boxes	Mould retains its shape during handling and metal pouring
Cohesiveness / Strength	Ability of sand particles to stick together	Mould retains desired shape even when hot metal is poured in
Refractoriness	Ability to withstand high temperatures of molten metal without fusing	Poor refractoriness causes sand to burn onto casting. Measured by sinter point.
Collapsibility	Ability of mould to collapse after casting solidifies	Facilitates easy withdrawal of casting; prevents hot tearing
Flowability	Ability to behave like fluid — flow to all mould portions and around pattern	Ensures good impression of pattern; increases with clay and water content

 **Most asked property question:** Properties of moulding sand appear in almost every exam. Memorise all 6 properties with their definitions. The exam accepts "any 4" usually — know all 6.

FACTORS AFFECTING PROPERTIES OF MOULDING SAND

- **Permeability:** Affected by — grain size, shape, moulding method, moisture content, clay content
- **Strength — Three types:**
 - **Green Strength:** Strength when mould has moisture (Primary for size/shape retention during handling)
 - **Dry Strength:** Strength when mould has no moisture
 - **Hot Strength:** Strength at temperature of molten metal (important for carrying weight of molten metal)
- **Refractoriness:** Depends on purity of sand, grain size, type of clay used
- **Collapsibility:** Important after casting solidification. Facilitates easy withdrawal.

 PREVIOUS YEAR QUESTIONS — UNIT 2 (MOULDING SAND)

MTE SEP 20254 marks · CO1

Q5. Describe the Properties of moulding sand. (Any four properties are sufficient)

- **Porosity (Permeability):** Sand must be porous to allow gases/moisture present within the mould to be removed freely when molten metal is poured.
- **Adhesiveness:** Sand particles must be capable of adhering to sides of moulding boxes.
- **Cohesiveness/Strength:** Ability of sand particles to stick together so that mould can be formed to desired shape even after hot metal is poured.
- **Refractoriness:** Moulding sands with poor refractoriness may burn onto the casting. Measured by sinter point of sand.
- **Collapsibility:** Mould must be able to collapse as the molten metal solidifies in it.
- **Flowability:** Ability of sand to behave like a fluid so that when rammed, sand flows to all portions of a mould and around the pattern, taking up required shape.

 EXPECTED/SAMPLE QUESTIONS — UNIT 2

 WARM

1. Differentiate between green sand and dry sand. (2 marks)

 LIKELY

2. What is the role of clay and water in green sand moulding? (2 marks)

 WARM

3. Explain green strength, dry strength, and hot strength of moulding sand. (4 marks)

 HOT

4. Why is permeability important in moulding sand? What factors affect it? (4 marks)

 LIKELY

5. What is the purpose of parting sand? Where is it used? (2 marks)

| GREEN SAND MOULD MAKING — STEP BY STEP

1. Place pattern on moulding board. Place drag on the board.
2. Sprinkle dry facing sand over board and pattern for non-sticky layer.
3. Riddle moulding sand in to cover pattern, then fill drag completely.
4. Firmly pack sand with hand rammers — neither too hard nor too soft.
5. Level off excess sand with strike-off bar.
6. Make vent holes with vent rod to facilitate gas removal during pouring.
7. Roll over finished drag flask exposing the pattern.
8. Place cope half of pattern over drag with locating pins. Place cope flask on drag.
9. Sprinkle dry parting sand over drag and pattern.
10. Place sprue pin and riser pin at appropriate places.
11. Fill, ram and vent the cope in the same manner as drag.
12. Remove sprue and riser pins. Scoop out pouring basin at top.
13. Remove pattern from cope and drag. Apply facing sand paste on mould cavity and runners.
14. Assemble the mould — ready for pouring.

| CORE

A **core** is a specialized insert, typically made of sand, that creates **hollow spaces or intricate internal features** within a cast part. Cores create internal cavities, recesses, and undercuts that the mold itself cannot form.

Example: Engine blocks with internal cooling passages need cores.

Core Making Steps

1. **Selection of Sand:** Core sand needs permeability, green strength, refractoriness, flowability, collapsibility, adhesiveness, cohesiveness, dry strength
2. **Mixing:** Sand mixed with binder (linseed oil, phenolic resin), moisture, and additives (coal dust, dextrin, silica flour)
3. **Blowing/Shooting:** Mixture blown into core box using core blowing machines
4. **Venting:** Vent passages made to allow gas escape during casting
5. **Reinforcement:** Heavy cores reinforced with cast iron grids or low-carbon steel wires
6. **Baking:** Core baked in oven at 200–250°C. Heat polymerizes the binder, giving strength.
7. **Cleaning & Finishing:** Trimming, brushing, and coating with refractory material
8. **Inspection:** Manual or machine inspection for dimensional accuracy
9. **Assembly:** Cores assembled with core glue and placed in mould

| GATING SYSTEM

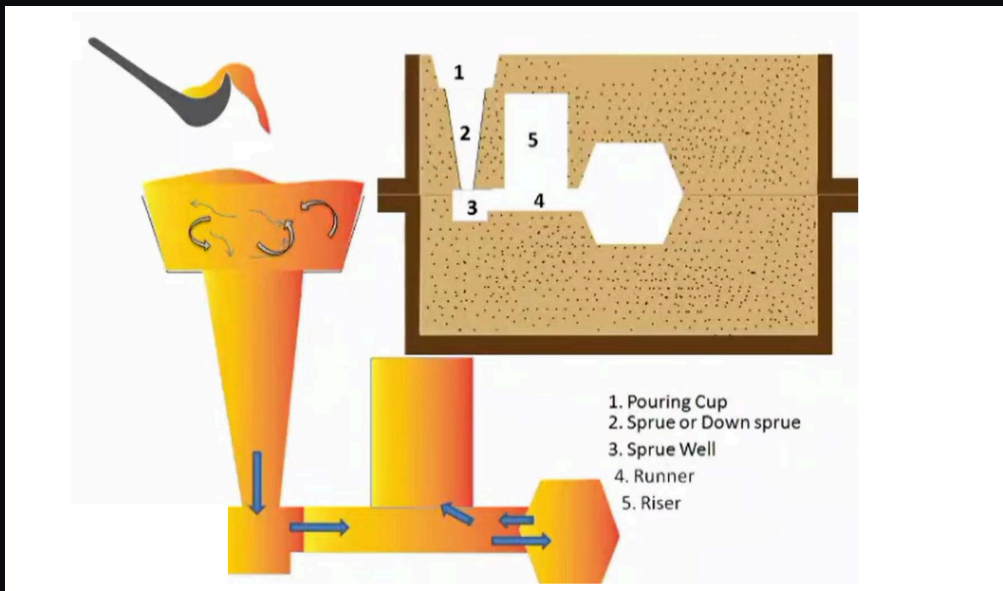


Fig: Gating system showing pouring basin, sprue, runner, gate, riser and casting cavity

The gating system is the network of channels through which molten metal flows from the ladle into the mould cavity.

Functions

- Controls flow rate of molten metal — prevents turbulence
- Minimizes defects — prevents air entrapment, shrinkage, porosity
- Directs metal flow to specific areas of complex shapes
- Riser provides shrinkage compensation

Types of Gates

GATE TYPE	METAL ENTRY	ADVANTAGES / DISADVANTAGES
Top Gate	From top of mould cavity	Simple; but causes turbulence and erosion of mould
Bottom Gate	From bottom of mould cavity	Smooth fill; but temperature gradient — bottom hotter, top cooler
Parting Line Gate	At the parting line where mould halves meet	Simple and cost-effective; prone to erosion and dross formation
Step Gate	Multiple entry points in steps	For heavy, large castings requiring large metal volume

 **Aspiration effect:** If pressure anywhere in the liquid metal stream becomes negative, outside gases (from burning organics in mould) get sucked in, causing porous castings. Good gating design avoids this.

 **PREVIOUS YEAR QUESTIONS — UNIT 3 (MOULD MAKING, CORE & GATING)**

ETE NOV 20256 marks · CO1

QB1. Prepare a diagram of gating system used in casting and explain function of its various components.

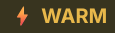
See detailed answer in Unit 1 PYQ section above (QB1). Draw the diagram showing: Pouring Basin → Sprue → Sprue Well → Runner → Runner Extension → Gate → Mould Cavity, with Riser branching off.

ETE NOV 202510 marks · CO3

QD1. Give any 5 parts of Lathe machine and give their functions in detail.

See Unit 14 — Lathe for full answer.

 **EXPECTED/SAMPLE QUESTIONS — UNIT 3**

 **WARM**

1. What is a core? Why is it used in casting? Give an example. (2 marks)

 **WARM**

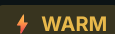
2. Explain the steps involved in core making. (6 marks)

 **HOT**

3. What are the requirements of a good gating system? (4 marks)

 **LIKELY**

4. Differentiate between a gate and a riser. (2 marks)

 **WARM**

5. Describe green sand mould making process with step-by-step procedure. (6 marks)

UNIT 4

Casting Methods

I

DIE CASTING

Die casting uses **high pressure** to force molten metal into a metal mould (die). Two main types:

HOT CHAMBER DIE CASTING

- For **low melting point metals** (~650°C)
- Metals: Lead, Tin, Zinc
- Metal starts in heated cylinder
- Piston forces metal into die
- Faster than cold chamber; short cycle times

COLD CHAMBER DIE CASTING

- For **high melting point metals** (>700°C)
- Metals: Copper, Brass, Aluminium
- Metal heated in separate crucible
- Ladled into cold chamber, then forced into die
- Higher pressures used

Advantages of Die Casting

- Extremely smooth surfaces (≈1 μm)
- Excellent dimensional accuracy
- Rapid production rate
- Better mechanical properties than sand casting
- Intricate parts possible

Disadvantages

- High initial die cost
- Limited to high-fluidity non-ferrous metals

- Part size limited
- Porosity may be a problem

Applications

Carburetors, automotive parts, bathroom fixtures, toys — using aluminium, zinc, magnesium, lead alloys.

| CENTRIFUGAL CASTING

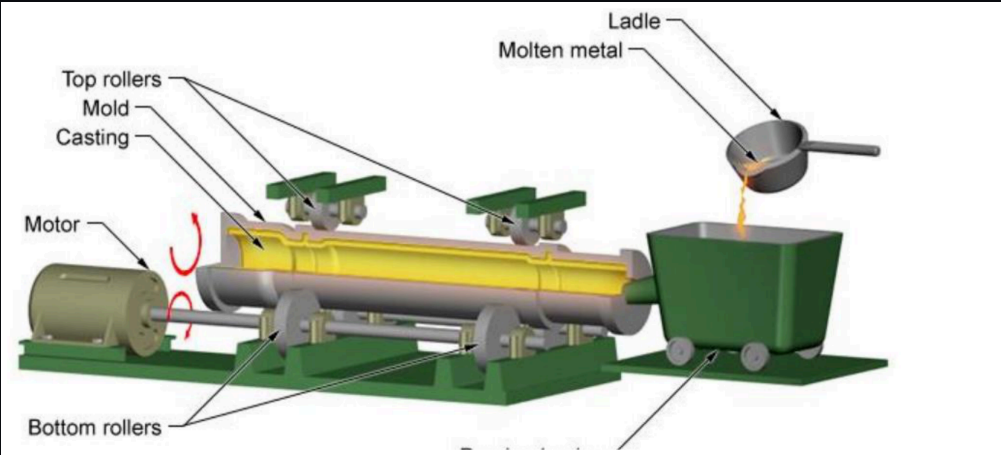


Fig: Horizontal centrifugal casting setup with rotating mould, pouring basin and sand lining

Uses **centrifugal force** to distribute molten metal into a rotating mould. Used for cylindrical/symmetrical parts.

Process

1. Mould set up and rotated (vertical: moderate RPM; horizontal: 200–1000 RPM)
2. Mould coated with refractory coating
3. Molten metal poured into rotating mould
4. Centrifugal force holds metal against mould wall until solidification
5. Lower-density impurities migrate toward center of rotation

Products: Pipes, tubes, bushings, cylinder liners, rings

Advantages: Uniform metal distribution, no core needed for hollow cylinders, impurities concentrate at center (machined away), dense outer structure

| INVESTMENT CASTING (LOST-WAX CASTING)

A versatile process for creating intricate, high-precision parts with complex geometries and smooth finishes.

Process Steps

1. **Pattern Creation:** Wax patterns injection-moulded in metal die. Patterns assembled into "pattern tree".
2. **Mould Creation:** Pattern tree dipped in fine ceramic slurry, coated with coarse particles, dried. Repeated to build thick ceramic shell.
3. **Lost Wax:** Shell placed in oven — wax melts out. Hollow ceramic shell = one-piece mould.
4. **Pouring:** Mould preheated to ~1000°C. Molten metal poured in.
5. **Cooling:** Metal solidifies in mould shape.
6. **Casting Removal:** Ceramic mould broken (water jets, cold breaking). Parts cut from gating system.
7. **Finishing:** Grinding, sandblasting, heat treatment if needed.

Materials: Alloy steel, carbon steel, stainless steel, aluminium, titanium, nickel, copper

Part size: 0.02 oz to 500 lb

Applications: Turbine blades, medical devices, jewellery, aerospace components, dental implants



PREVIOUS YEAR QUESTIONS — UNIT 4 (CASTING METHODS)

MTE SEP 2025

4 marks · CO1

Q7. Describe centrifugal casting method.

Full answer provided in Unit 1 PYQs above. Key points: rotating mould, centrifugal force distributes molten metal, low-density impurities rise to centre, produces cylindrical/symmetrical parts like pipes and tubes.

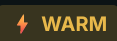
ETE NOV 2025 (IMPLIED)

4 marks · CO1

Differentiate between machining process and additive manufacturing process.

See Unit 20 — Additive Manufacturing for complete comparison table.

💡 EXPECTED/SAMPLE QUESTIONS — UNIT 4

-  **HOT** 1. Differentiate between hot chamber and cold chamber die casting. (4 marks)
-  **WARM** 2. Describe investment (lost-wax) casting process with steps. (6 marks)
-  **HOT** 3. What are the advantages of centrifugal casting? (2 marks)
-  **LIKELY** 4. For which metals is hot chamber die casting used? Give reasons. (2 marks)
-  **LIKELY** 5. Why is the ceramic mould in investment casting preheated before pouring? (2 marks)

UNIT 5

Welding — Importance, Concepts & Classification

| WHAT IS WELDING?

Welding is a materials joining process in which two or more parts are coalesced (fused together) at their contacting surfaces by suitable application of heat and/or pressure. The assembly of joined parts is called a **weldment**.

- Some processes use **heat only** (e.g., arc welding)
- Some require **both heat and pressure** (e.g., resistance welding)
- Some use **pressure only, no heat** (e.g., cold welding)
- Some require a **filler material** to facilitate coalescence

| ENERGY SOURCES IN WELDING

ENERGY SOURCE	WELDING PROCESSES
Chemical Energy	Gas welding, Explosive welding, Thermit welding
Mechanical Energy	Friction welding, Ultrasonic welding
Electrical Energy	Arc welding, Resistance welding
Radiation Energy	Laser beam welding (LBM), Electron beam welding (EBM)

| ZONES AT THE WELD AREA

- **Fusion Zone:** Mixture of filler metal and base metal completely melted; high homogeneity
- **Weld Interface Zone:** Narrow boundary between fusion zone and HAZ; thin band that melted and immediately resolidified
- **Heat-Affected Zone (HAZ):** Metal below melting point but high enough to cause microstructural changes in solid metal

| TYPES OF WELDING JOINTS

- **Butt Joint** — two pieces aligned end-to-end
- **Lap Joint** — one piece overlaps another
- **T-Joint** — one piece perpendicular to another
- **Corner Joint** — two pieces at right angles at their edges
- **Edge Joint** — edges of two pieces joined

| COMMON WELDING DEFECTS

DEFECT	CAUSE	REMEDY
Porosity	Release of trapped gases during solidification	Clean surfaces before welding; remove contamination
Undercut	High current, improper electrode angle, long arc length	Appropriate current, use smaller electrode
Slag Inclusion	Insufficient current, slow travel speed	Proper current, appropriate travel speed; remove by grinding
Incomplete Fusion	Low current, fast travel speed, short arc gap	Appropriate current and travel speed; remove and reweld
Weld Spatter	High arc current, magnetic arc blow	Reduce arc current, shorten arc length; remove by grinding
Incomplete Penetration	Low current, fast travel speed, short arc length	Appropriate parameters; clean and reweld



QA2. Give one prime difference between welding and brazing.

Welding: Involves direct melting of the base metals being joined. The welding alloy has a higher melting point than the metals joined.

Brazing: Does NOT involve melting of base metals. Only the filler (brazing alloy) melts. Filler melting point is above 450°C but below melting point of base metals.

Q2. What is the function of shielding gas in welding?

Shielding gas (e.g., Argon, CO₂, Helium) protects the molten weld pool from atmospheric contamination — particularly oxygen and nitrogen — which would cause oxidation, porosity, and weakened joints. It creates a protective atmosphere around the arc and weld pool.

💡 EXPECTED/SAMPLE QUESTIONS — UNIT 5

- ⚡ WARM1. What is welding? Classify welding processes based on energy source. (4 marks)
- 💧 LIKELY2. Explain the three zones of a welded joint. (3 marks)
- 💧 LIKELY3. What is the heat-affected zone (HAZ) in welding? (2 marks)
- ⚡ WARM4. List five common welding defects and their causes. (5 marks)
- 🔥 HOT5. What is flux in welding? What is its function? (2 marks) [MTE Q2]

UNIT 6

Gas Welding & Types of Flames

I GAS WELDING EQUIPMENT

- Gas cylinders (Oxygen — black; Acetylene — maroon/red)
- Pressure regulators — maintain supply pressure at constant value
- Pressure gauges — one for cylinder pressure, one for supply pressure
- Hose pipes and valves — colour-coded same as cylinders; made of rubber
- Blow pipe (torch) and spark lighter
- Welding screens, goggles, gloves, apron

💡 Oxygen is at 2000 PSI in black cylinder. Acetylene is dissolved in acetone in maroon cylinder. Cylinders have different thread types (Left hand & Right hand) for safety — cannot be accidentally swapped.

I OXY-ACETYLENE WELDING

Heat generated by chemical reaction of oxygen + acetylene. Temperature of flame: approximately **3300°C**. Both gases mixed in chamber of torch and ignited.

I TYPES OF GAS WELDING FLAMES

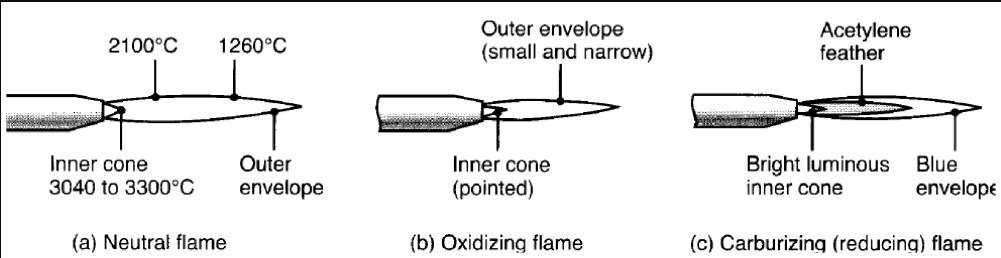


Fig: Neutral, oxidizing and carburizing oxy-acetylene flames with flame zones

FLAME TYPE	OXYGEN:ACETYLENE RATIO	APPEARANCE	APPLICATIONS
Neutral Flame	1:1 (equal amounts)	Two zones: inner white cone + outer blue envelope. No excess of either gas.	Most widely used. Welding of steel, stainless steel, copper, cast iron. Does not oxidize or carburize the metal.
Carburizing Flame (Reducing)	Excess Acetylene	Three zones: inner cone + middle acetylene feather + outer envelope. Feather size increases with more acetylene.	Welding of high-carbon steel, nickel alloys, hard-facing. Adds carbon to weld metal.

FLAME TYPE	OXYGEN:ACETYLENE RATIO	APPEARANCE	APPLICATIONS
Oxidizing Flame	1.15–1.50:1 (Excess Oxygen)	Inner cone shorter than neutral; light blue colour; burns with harsh hissing sound.	Welding of brass, bronze. Oxidizes weld metal — avoid for steel.

 **Exam favourite:** "Describe various types of flames used in gas welding with neat sketches." Appears in Re-sessional Apr 2026 (Q4, 4 marks). Draw all 3 flames clearly showing zones.

I GAS CUTTING

- Metal preheated by oxy-fuel flame to ~900°C
- Pure oxygen jet directed at hot metal causes exothermic oxidation
- Iron oxide (fluid slag) formed and blown away by oxygen jet
- Moving torch cuts a continuous slot
- Can cut steel sections up to 300 mm thick

 **PREVIOUS YEAR QUESTIONS — UNIT 6 (GAS WELDING & FLAMES)**

RE-SESSIONAL APR 2026

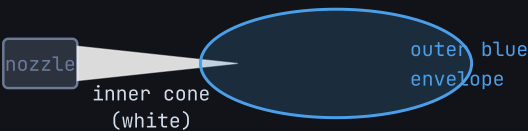
4 marks · CO3

Q4. Describe various types of flames used in gas welding with neat sketches.

Three types of oxy-acetylene flames:

NEUTRAL FLAME

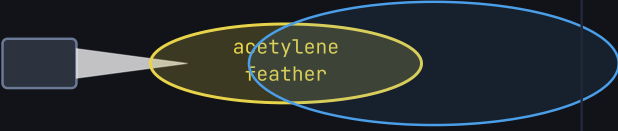
O₂ : C₂H₂ = 1:1



✓ 2 zones only – balanced
Best for: Steel, Cu, Cast Iron

CARBURIZING FLAME

Excess Acetylene – 3 zones



✓ 3 zones – adds carbon to weld
Best for: High-C steel, hard-facing

OXIDIZING FLAME

Excess O₂ – ratio 1.15-1.5:1



X Avoid for steel – oxidises we
Best for: Brass, Bronze only

Oxy-Acetylene Flame Types – Exam diagram (draw all 3 zones for carburizing)

Fig: Three types of oxy-acetylene flames — Neutral (2 zones), Carburizing (3 zones with feather), Oxidizing (short cone, harsh sound)

- **Neutral Flame:** Equal O₂ and C₂H₂. Two zones — white inner cone + blue outer envelope. Used for welding steel, copper, cast iron. Does not add or remove carbon.
- **Carburizing Flame:** Excess acetylene. Three zones — inner cone + intermediate acetylene feather + outer envelope. Adds carbon to weld. Used for high-carbon steel, hard-facing.
- **Oxidizing Flame:** Excess oxygen (ratio 1.15–1.5:1). Inner cone shorter and blue. Burns with harsh sound. Used for brass/bronze. Oxidizes the weld pool.

 **EXPECTED/SAMPLE QUESTIONS — UNIT 6**

 **WARM**

1. With a neat sketch, describe the oxy-acetylene welding equipment. (4 marks)

 **LIKELY**

2. Why are oxygen and acetylene cylinders given different thread types? (2 marks)

 **HOT**

3. Which flame is used for welding steel? Explain its characteristics. (2 marks)

 **LIKELY**

4. What is gas cutting? How does it differ from gas welding? (3 marks)

 **WARM**

5. Why is oxidizing flame avoided for steel welding? (2 marks)

UNIT 7

Electric Arc Welding & Submerged Arc Welding

I INTRODUCTION TO ARC WELDING (SMAW)

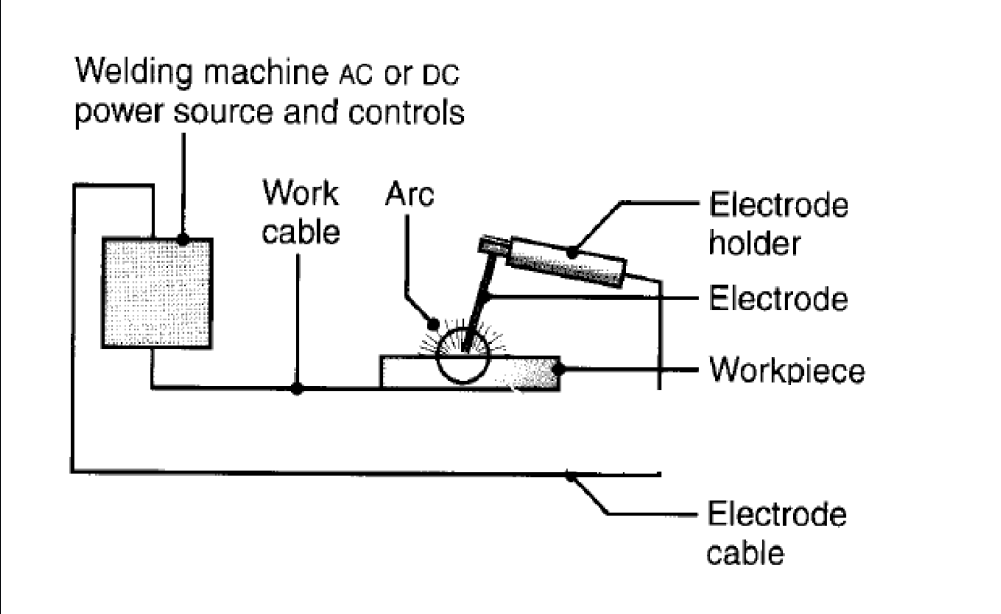


Fig: Basic arc welding circuit showing power source, work cable, electrode holder and arc

Shielded Metal Arc Welding (SMAW) — also called Manual Arc Welding. Simple, portable, inexpensive. Suitable for almost all metals. Low initial cost.

SMAW Equipment

- Welding machine/transformer (AC or DC)
- Electrode holder, cables
- Flux-coated electrode
- Chipping hammer, wire brush
- Welding shield (UV protection), goggles, gloves, apron

AC vs DC in Welding

PARAMETER	AC	DC
Voltage required	80–110 V	8–25 V
Current	50–300 A	50 A typical
Best for	Ferrous metals	Non-ferrous and special metals
Machine type	Transformer (converts supply to low V, high A)	Motor-generator or rectifier

Polarity in DC Welding

- **Straight Polarity (DCEN):** Electrode = negative, Workpiece = positive. Two-thirds of heat at workpiece. More penetration.
- **Reversed Polarity (DCEP):** Electrode = positive, Workpiece = negative. More heat at electrode. Used for thin sheets to avoid burn-through.

Electrodes

- **Consumable electrodes:** Provide filler metal. Types: coated (with flux) and bare.
- **Non-consumable electrodes:** GTAW (TIG welding) — additional filler wire required.
- **Bare electrodes:** Used in gas-shielded welding (GMAW/MIG)

Functions of Flux Coating on Electrodes

- Develops reducing atmosphere and prevents oxidation
- Forms separable slag from metal impurities
- Establishes arc stability
- Improves bead appearance
- Avoids weld spatter
- Provides alloying elements to weld pool

I SUBMERGED ARC WELDING (SAW)

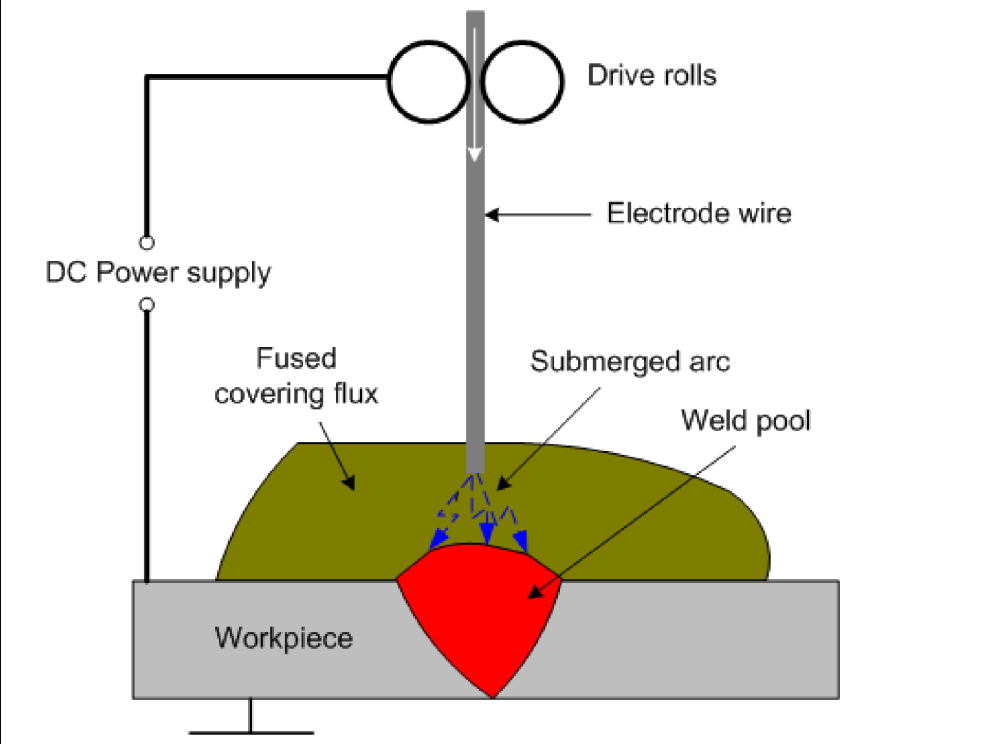


Fig: Submerged arc welding showing electrode wire, flux layer, submerged arc and weld pool

The arc and molten metal are completely submerged under a blanket of granular flux — hence "submerged."

Process

- Flux (Calcium Oxide, Calcium Fluoride, Silica — sintered powder) spread on joint
- Metal electrode fed automatically into flux
- Arc forms under flux blanket — not visible
- Flux melts and covers weld bead
- AC or DC used; current: 200–2000 amperes
- Used to weld thick plates

Advantages: High deposition rates, deep penetration, excellent surface finish, no UV radiation (submerged), suitable for automation

PREVIOUS YEAR QUESTIONS — UNIT 7 (ARC WELDING)

ETE NOV 2025 6 marks · CO2

QB2. Describe the process of submerged arc welding.

Submerged Arc Welding (SAW):

- Granular flux (Calcium Oxide, CaF₂, Silica) is spread over the joint to be welded
- A bare metal electrode is fed automatically into the flux
- Electric arc forms between electrode and workpiece — completely submerged under flux
- Flux melts and forms a protective slag layer over the weld bead
- Unmelted flux is recovered and reused
- Current rating: 200–2000 A (AC or DC)
- Used for welding thick plates in shipbuilding, pressure vessels, structural fabrication

Advantages: High deposition rate, deep penetration, smooth bead, no harmful UV arc exposure, high automation potential, high quality welds.

Disadvantages: Limited to flat/horizontal positions, flux handling required, not visible arc.

EXPECTED/SAMPLE QUESTIONS — UNIT 7

- ⚡ WARM

1. Explain straight polarity and reversed polarity in DC arc welding. (4 marks)
- ⚡ WARM

2. What are the functions of flux coating on SMAW electrodes? (4 marks)
- 💧 LIKELY

3. What is SMAW? Give its advantages. (2 marks)
- 🔥 HOT

4. With a neat sketch, explain the submerged arc welding process. (6 marks)
- 💧 LIKELY

5. What is the difference between consumable and non-consumable electrodes? (2 marks)

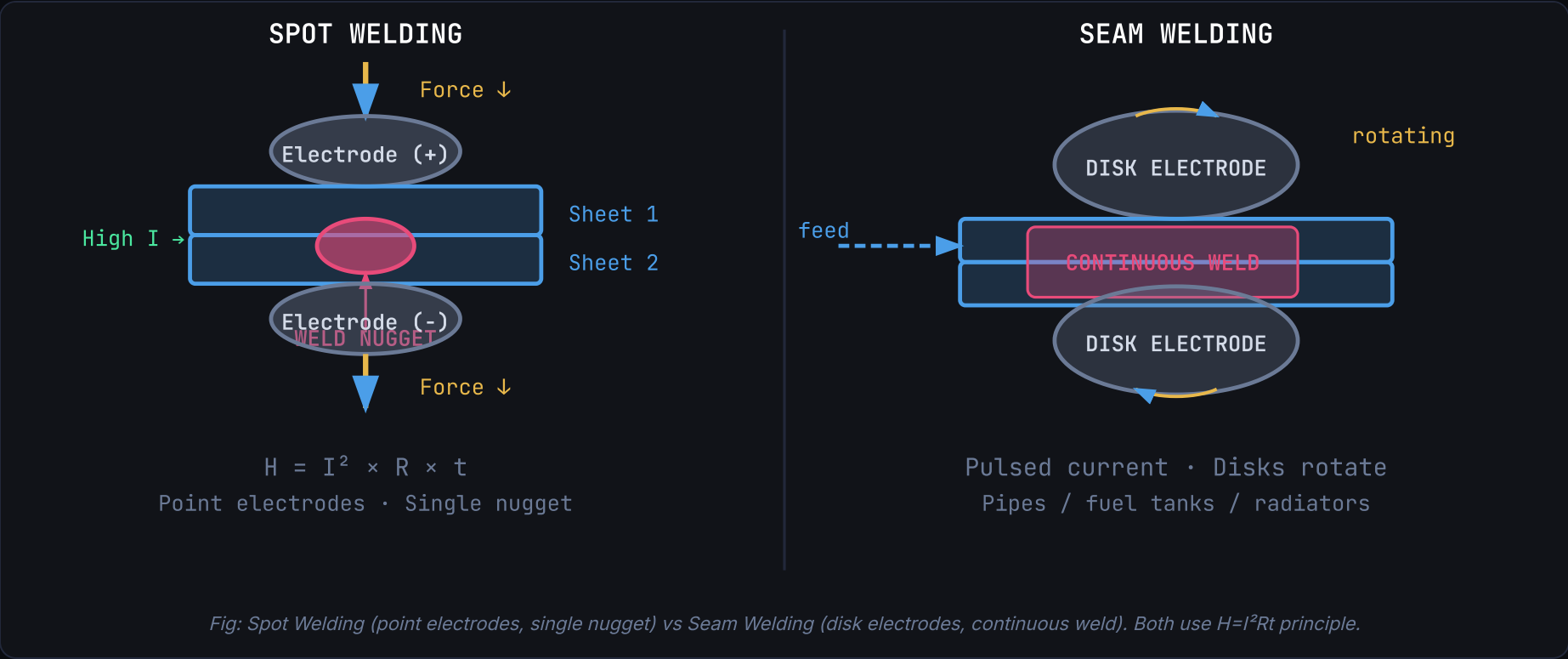
UNIT 8 Resistance Welding

Heat is generated due to **electrical resistance** when current flows through the joint. Formula: **H = I²Rt**, where H = heat, I = current, R = resistance, t = time. Pressure is also applied simultaneously.

I TYPES OF RESISTANCE WELDING

1. Spot Welding

- Two overlapping sheets clamped between two copper electrodes
- High current passed for short time — creates a single "spot" weld nugget
- Electrodes apply pressure before, during, and after current flow
- Applications: Automotive body panels, sheet metal assemblies



2. Seam Welding

- Disk/roller-type copper electrodes (instead of point electrodes)
- Produces a **continuous** weld along a seam
- Current applied in pulse form at time intervals
- Disks rotate continuously — metal sheets move forward
- Applications: Seam-welded pipes, fuel tanks, radiators

3. Projection Welding

- Modification of spot welding
- One sheet has circular projections; other is flat
- Projection diameter = plate thickness
- Current and pressure concentrated at projection areas
- Upper electrode movable; lower fixed
- Projections melt and weld is made
- Can make multiple welds simultaneously

4. Flash Butt Welding

- Used for **butt joints**
- Two surfaces brought into contact or near contact
- Electric current heats surfaces to melting point
- Arcing occurs at faying surfaces (called "flashing") — hence the name
- Surfaces then forced together to form weld
- Current stopped during upsetting (pressing together)
- Sometimes classified under arc welding due to flashing

5. Upset Butt Welding

- Similar to flash welding but faying surfaces are pressed together *during* heating
- No arcing — heating entirely by electrical resistance at contact
- When surfaces reach suitable temperature, force increased to cause upsetting and coalescence
- Not a fusion-welding process in the traditional sense

Q8. Describe any four methods of resistance welding.

1. **Spot Welding:** Two copper rod electrodes press overlapping sheets together. High current passed — creates weld nugget at contact point. Used in automotive body manufacturing.
2. **Seam Welding:** Disk-type copper electrodes rotate continuously moving metal sheets forward. Current applied in pulses. Produces continuous weld bead. Used to manufacture seam-welded pipes and water-tight joints.
3. **Projection Welding:** Modification of spot welding. One sheet has circular projections (diameter = plate thickness). Sheets kept between copper electrodes. Current and pressure concentrated at projections — they melt, forming welds simultaneously.
4. **Flash Butt Welding:** Two surfaces brought to near contact. Current applied → surfaces heat to melting point with arcing (flashing). Surfaces then forced together forcefully — forming butt weld. Current stopped during upsetting.

ETE NOV 2025 (IMPLIED) 2 marks · CO2

Name any two cutting tool materials.

High Speed Steel (HSS) and Cemented Carbide (Tungsten Carbide). (See Unit 13 for full list)

 EXPECTED/SAMPLE QUESTIONS — UNIT 8

-  **HOT** 1. Explain the principle of resistance welding. (2 marks)
-  **WARM** 2. How does seam welding differ from spot welding? (4 marks)
-  **WARM** 3. What is projection welding? How is it different from spot welding? (3 marks)
-  **HOT** 4. Explain flash butt welding and upset butt welding. (4 marks)
-  **LIKELY** 5. What type of electrodes are used in seam welding and why? (2 marks)

UNIT 9

Soldering & Brazing

| SOLDERING

A method of uniting two metal pieces using a **dissimilar metal or alloy** (solder) by application of heat. Solder melting point is **below 450°C**. Base metals are NOT melted.

Types of Solder

- **Soft Solder:** Alloy of Lead (37%) + Tin (63%). Melting temperature: 150–350°C. Common electronic soldering.
- **Hard Solder:** Alloy of Copper + Tin + Silver. Soldering temperature: 600–900°C. Used for stronger joints.
- **Flux used:** Zinc Chloride — cleans joint surfaces and prevents oxidation during heating.

Soldering Process Steps

1. **Preparation:** Clean metal surfaces — remove oil, dirt, oxidation with solvent or sandpaper
2. **Apply Flux:** Apply flux paste to joint area to remove oxide layers and prevent new formation
3. **Heat the Joint:** Touch soldering iron tip to joint surfaces (heat the joint, not the solder)
4. **Apply Solder:** Bring solder wire into contact with heated joint — heat from joint melts solder, which flows in
5. **Finish:** Remove iron, allow joint to cool and solidify, clean flux residue

| BRAZING

A method of joining two similar or dissimilar metals using a **fusible filler alloy**. Brazing temperature is above 450°C (typically 450–1200°C) but **below the melting points of the base metals**. Filler spreads by **capillary action**.

Brazing Process Steps

1. **Cleaning:** Remove oxides, grease, contamination
2. **Joint Design:** Precise small gap (0.05–0.2 mm) to facilitate capillary action
3. **Fluxing:** Apply flux to prevent oxidation and promote wetting
4. **Heating:** Heat above filler metal melting point but below base metal melting point
5. **Apply Filler Metal:** Filler drawn into gap by capillary action
6. **Cooling:** Solidification creates permanent metallurgical bond
7. **Post-Brazing:** Clean flux residue

Types of Brazing (Heat Sources)

- **Torch Brazing:** Oxy-acetylene or propane torch
- **Induction Brazing:** Induction coil generates heat
- **Furnace Brazing:** Assembly placed in furnace for even heating
- **Resistance Brazing:** Electrical resistance creates heat at joint

| COMPARISON OF SOLDERING, BRAZING, AND WELDING

FEATURE	SOLDERING	BRAZING	WELDING
Filler melting point	Below 450°C	450°C – 1200°C	High (above base metal)
Base metal melted?	No	No	Yes (direct melting)
Joint strength	Weaker	Stronger than soldering	Strongest
Surface finish	Does not yield good finish	Good surface finish	Depends on process
Similar/dissimilar metals	Preferably similar	Dissimilar easily	Usually similar

 PREVIOUS YEAR QUESTIONS — UNIT 9 (SOLDERING & BRAZING)

ETE NOV 20252 marks · CO2

QA2. Give one prime difference between welding and brazing.

In **welding**, the base metals are directly melted and fused. In **brazing**, the base metals are NOT melted — only the filler alloy melts and flows into the joint by capillary action at temperatures above 450°C.

 EXPECTED/SAMPLE QUESTIONS — UNIT 9

-  **HOT** 1. Compare brazing and soldering on any four aspects. (4 marks)
-  **WARM** 2. What is flux? Why is it used in soldering and brazing? (2 marks)
-  **WARM** 3. Explain the brazing process step-by-step. (4 marks)
-  **LIKELY** 4. What is capillary action in brazing? Why is a small gap (0.05–0.2mm) required? (2 marks)
-  **LIKELY** 5. Give the composition of soft solder and its melting temperature. (2 marks)

UNIT 10

Metal Forming: Hot & Cold Working

| WHAT IS METAL FORMING?

Metal Forming is a fabrication process in which parts are produced by subjecting the material to heavy **plastic deformation**. The material changes shape permanently without material removal (unlike machining).

Plastic deformation = permanent change in shape when a force is applied beyond the elastic limit (yield point).

| CLASSIFICATION OF METAL FORMING

BULK FORMING

- Severe deformation; significant shape change
- Small surface area-to-volume ratio
- Mostly done in hot working conditions
- **Processes:** Rolling, Forging, Extrusion, Wire Drawing

SHEET METAL FORMING

- Forming of thin sheets and coils
- High surface area-to-volume ratio
- Uses punches and dies
- **Processes:** Shearing, Bending, Drawing, Blanking, Punching

| HOT WORKING VS COLD WORKING

ASPECT	HOT WORKING	COLD WORKING
Temperature	Above recrystallization temperature (0.5–0.75 Tm)	Below recrystallization temperature (room temp)
Force Required	Less (metal is soft and ductile)	More (metal is hard and strong)
Strength & Hardness	Low (no strain hardening; recrystallization occurs)	High (due to strain hardening)

ASPECT	HOT WORKING	COLD WORKING
Ductility	High	Reduced
Surface Finish	Poor (oxidation, scale formation)	Good (smooth, bright surface)
Dimensional Accuracy	Low	High
Residual Stresses	Not developed	Developed in material
Energy Consumption	Lower (soft metal deforms easily)	Higher (harder metal resists deformation)
Grain structure	New strain-free crystals formed	Distorted grain structure

 **Most asked question:** Hot vs Cold working comparison table appears in MTE Sep 2025 (Q4, 4 marks). Know at least 6–8 differences for full marks.

Hot Working Advantages & Disadvantages

ADVANTAGES

- Significant plastic deformation possible
- Significant shape change
- Lower forces required
- Materials with premature failure can be hot formed
- No work hardening (absence of strain hardening)

DISADVANTAGES

- Shorter tool life
- Poor surface finish
- Lower dimensional accuracy
- Surface oxidation occurs

SHEET METAL FORMING OPERATIONS

OPERATION	DESCRIPTION
Shearing	Cutting along a straight line between two cutting edges. Inclined blade used to reduce force.
Blanking	Sheet metal cutting to separate a piece (called a <i>blank</i>) from surrounding stock. The <i>blank</i> is the desired part.
Punching/Piercing	Removes a scrap slug from workpiece each time punch enters die. Leaves a HOLE in the workpiece. The slug is scrap.
Bending (V-bending)	Sheet strained around a straight axis using V-shaped die. For low production on brake press.
Deep Drawing	Forming flat sheet into cup/box shaped part. Blank holder holds sheet; punch pushes into die.

 **Blanking vs Punching:** In **blanking**, the punched-out piece IS the product (blank). In **punching**, the punched-out piece (slug) is SCRAP — the hole left behind is the product.

 **PREVIOUS YEAR QUESTIONS — UNIT 10 (METAL FORMING)**

MTE SEP 2025

4 marks · CO3

Q4. Differentiate between cold working and hot working.

Full comparison table provided above in this unit. Include all 8 aspects for full marks (temperature, force, strength, ductility, surface finish, dimensional accuracy, residual stresses, energy).

MTE SEP 2025

4 marks · CO3

Q3. What is plastic deformation of metals?

Plastic deformation of metals is the **permanent change in shape** of a metal when a force is applied **beyond its elastic limit (yield point)**. Unlike elastic deformation (which is recoverable), plastic deformation is permanent. It forms the basis of all metal forming processes like rolling, forging, and extrusion.

MTE SEP 2025

4 marks · CO3

Q6. Compare Blanking process with Punching process of sheet metal forming.

Blanking: Sheet metal cutting to separate a piece called a *blank* from surrounding stock. The punched-out piece IS the desired part (product). The surrounding sheet becomes scrap.

Punching: A metal fabricating process that removes a scrap slug each time a punch enters the die. This process leaves a HOLE in the workpiece. The hole in the sheet is the product. The slug is scrap.

Key difference: In blanking, the blank is the part. In punching, the hole is the part.

RE-SESSIONAL NOV 2025 4 marks · CO3

Q3. Differentiate between wire drawing and extrusion processes.

ASPECT	WIRE DRAWING	EXTRUSION
Action	Wire/rod is pulled through die opening	Metal billet is pushed/compressed through die
Force type	Tensile force (pulling)	Compressive force (pushing)
Output	Reduced diameter wire/rod	Any cross-sectional shape (not just circular)
Temperature	Usually cold	Can be hot or cold
Shape variety	Primarily circular cross-section	Complex shapes possible (I-beams, tubes, etc.)

RE-SESSIONAL NOV 2025 4 marks · CO3

Q5. Differentiate between Drawing and Deep drawing processes.

Wire/Bar Drawing: A bulk forming process where wire, rod, or bar is pulled through a die hole to reduce its cross-sectional area. Produces long lengths of reduced-diameter material.

Deep Drawing: A sheet metal forming process where a flat blank is formed into a cup, box, or hollow shape by pressing with a punch into a die opening. Uses a blank holder to prevent wrinkling. Products: beverage cans, automobile panels, kitchen sinks.

Key difference: Wire drawing reduces cross-section of a solid rod. Deep drawing creates hollow 3D shapes from flat sheet metal.

💡 EXPECTED/SAMPLE QUESTIONS — UNIT 10

- ⚡ WARM

1. What is metal forming? Classify into bulk forming and sheet metal forming with examples. (4 marks)
- 💧 LIKELY

2. What is warm working? How does it differ from hot and cold working? (3 marks)
- ⚡ WARM

3. Describe the deep drawing process with a neat sketch. (4 marks)
- 💧 LIKELY

4. Why is a blank holder used in deep drawing? (2 marks)
- 💧 LIKELY

5. What is V-bending? Where is it used? (2 marks)

UNIT 11 Forging, Rolling, Extrusion & Wire/Tube Drawing

I ROLLING

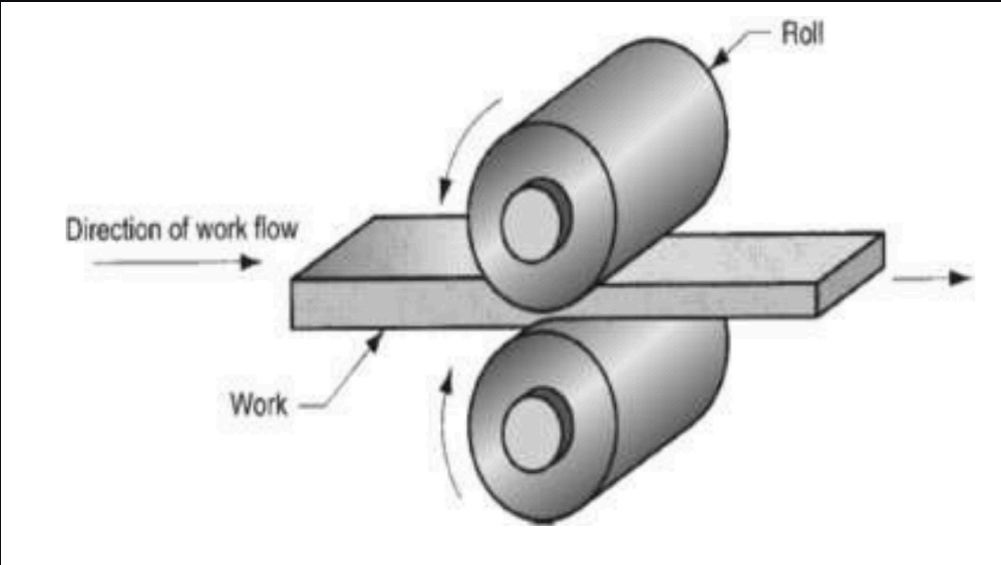


Fig: Rolling process showing workpiece thickness reduction between two rotating rolls

A metal forming process in which **thickness of work is reduced by compressive forces from two rolls rotating in opposite directions.**

Important Terminology

- **Bloom:** Square cross-section, ≥150×150 mm
- **Slab:** Rectangular cross-section, width ≥250 mm, thickness ≥40 mm

- **Billet:** Square cross-section, sides ≥40 mm

Types of Rolling Mills

MILL TYPE	DESCRIPTION	FEATURES
Two High Mill	Two rolls rotating in opposite directions. Reversing or non-reversing.	Roll diameter: 0.6–1.4 m; simple design
Three High Mill	Three rolls — top two for first pass, bottom two for further reduction	Direction cannot be reversed; needs mechanism to shift slab
Four High Mill	Two small working rolls + two large backing rolls	Small rolls reduce contact area and force; backing rolls prevent deflection
Tandem Mill	Series of 8–10 rolling stations; thickness reduced at each station	Work velocity increases with each station; continuous production
Cluster Mill	Small working rolls with multiple backing rolls	Very thin sheets possible; high accuracy

Applications of Rolling

- Blooms → structural shapes (rails for railroad tracks)
- Billets → bars, rods (raw material for machining, forging, extrusion)
- Slabs → plates, sheets, strips (shipbuilding, bridges, boilers)

| FORGING

A deformation process where workpiece is **compressed between two dies** using impact load (hammer) or hydraulic load (press).

Classification of Forging

BY TEMPERATURE	BY PRESS TYPE	BY DIE TYPE
Cold, Warm, Hot forging	Impact load → Hammer; Gradual → Press	Open die, Impression (closed) die, Flashless

Types of Forging Operations

- **Open Die Forging (Upsetting):** Billet compressed between flat dies. Diameter increases; height decreases. Simple shapes. Also called upset forging.
- **Impression Die (Closed Die) Forging:** Dies contain shaped impressions. Billet fills impression completely; excess forms "flash" at parting line. Multiple stages may be needed. Example: Coining.
- **Flashless Forging:** Volume of billet must exactly match die cavity. Complex, symmetrical shapes. Suitable for Al, Mg and their alloys.

Applications of Forging

Engine crankshafts, connecting rods, gears, aircraft structural components, jet engine turbine parts.

| EXTRUSION

A bulk forming process where work metal is **forced/compressed to flow through a die hole** to produce a desired cross-sectional shape. Like squeezing toothpaste from a tube.

Types of Extrusion

TYPE	DESCRIPTION	KEY DIFFERENCE
Direct (Forward) Extrusion	Ram pushes billet; metal flows through die in the same direction as ram motion	Higher friction (billet vs container wall); higher force needed
Indirect Extrusion	Die mounted on hollow ram; metal flows opposite to ram direction through die on ram side	No billet-container friction; lower force; but ram has lower rigidity
Impact Extrusion	High speed, short stroke impact. Can be forward, backward, or combined.	For thin-walled parts: toothpaste tubes, battery cases
Hydrostatic Extrusion	Billet surrounded by pressurized fluid; no container friction	Suitable for brittle materials; high reduction ratios; needs taper on billet end

| WIRE, ROD & BAR DRAWING

Wire, rod, or bar is **pulled through a die hole**, reducing cross-sectional area.

- **Bar Drawing:** Large diameter; single-stage; straight rods (not coiled)
- **Wire Drawing:** Small diameter (down to 0.03 mm); continuous process through 4–12 dies; wire drawn from coils; "continuous drawing"

| TUBE DRAWING

Reduces diameter or wall thickness of seamless tubes and pipes.

- **Tube Sinking (no mandrel):** Simplest; diameter reduced but inside diameter/thickness uncontrolled
- **Fixed Mandrel:** Mandrel attached to support bar; controls inside diameter and wall thickness; limits tube length
- **Floating Plug:** Mandrel floats inside tube; no length restriction; ideal position found naturally in reduction zone

 PREVIOUS YEAR QUESTIONS — UNIT 11 (FORMING PROCESSES)

ETE NOV 2025 10 marks · CO3

QC1. Explain metal rolling process, its features and applications.

Rolling Process: A metal forming process in which thickness of a workpiece is reduced by compressive forces exerted by two rolls rotating in opposite directions.

Features:

- The rotating rolls draw the slab into the gap and compress it
- Can be done hot (above recrystallization) or cold (below recrystallization)
- Types of mills: Two-high, Three-high, Four-high, Tandem, Cluster
- Products classified as: Bloom (square, ≥150mm), Slab (rectangular), Billet (square, smaller)

Applications:

- Blooms → Railroad rails, structural shapes
- Billets → Bars, rods for machining, drawing, forging
- Slabs → Plates for shipbuilding, bridges, boilers, heavy machinery
- Sheets/strips → Automotive body panels, appliances, packaging

 EXPECTED/SAMPLE QUESTIONS — UNIT 11

-  **WARM** 1. Explain open die and impression die forging with neat sketches. (6 marks)
-  **WARM** 2. What is direct extrusion? How does it differ from indirect extrusion? (4 marks)
-  **HOT** 3. Explain the two-high rolling mill. What is the difference between reversing and non-reversing mills? (4 marks)
-  **LIKELY** 4. What is tube sinking? When is a floating plug mandrel preferred? (3 marks)
-  **LIKELY** 5. Applications of forging: name 4 automobile components made by forging. (2 marks)

UNIT 12

Machining: Principles of Metal Cutting

| WHAT IS MACHINING?

Machining is a **subtractive process** used to obtain desired shape, size, and finish by **removing surplus material** in the form of chips by a cutting tool with suitable relative motion between workpiece and tool.

- **Cutting tool:** Removes excess material through direct mechanical contact
- **Machine tool:** Provides necessary relative motion between workpiece and tool

| TWO RELATIVE MOTIONS IN MACHINING

- **Cutting speed (Primary motion):** The main motion generating the cut. The line generated is called the *generatrix*.
- **Feed speed (Secondary motion):** Determines the area being machined. The line generated is called the *directrix*.

| TYPES OF MACHINING

ORTHOGONAL MACHINING

- Cutting edge perpendicular to cutting velocity
- Cutting forces in x and z directions only (2D)
- Examples: Sawing, Broaching, Parting off tube walls

OBLIQUE MACHINING

- Cutting edge inclined to cutting velocity direction
- Forces in all three directions (3D)
- Examples: Turning, Milling, Drilling, Shaping

| TYPES OF CHIPS IN METAL CUTTING

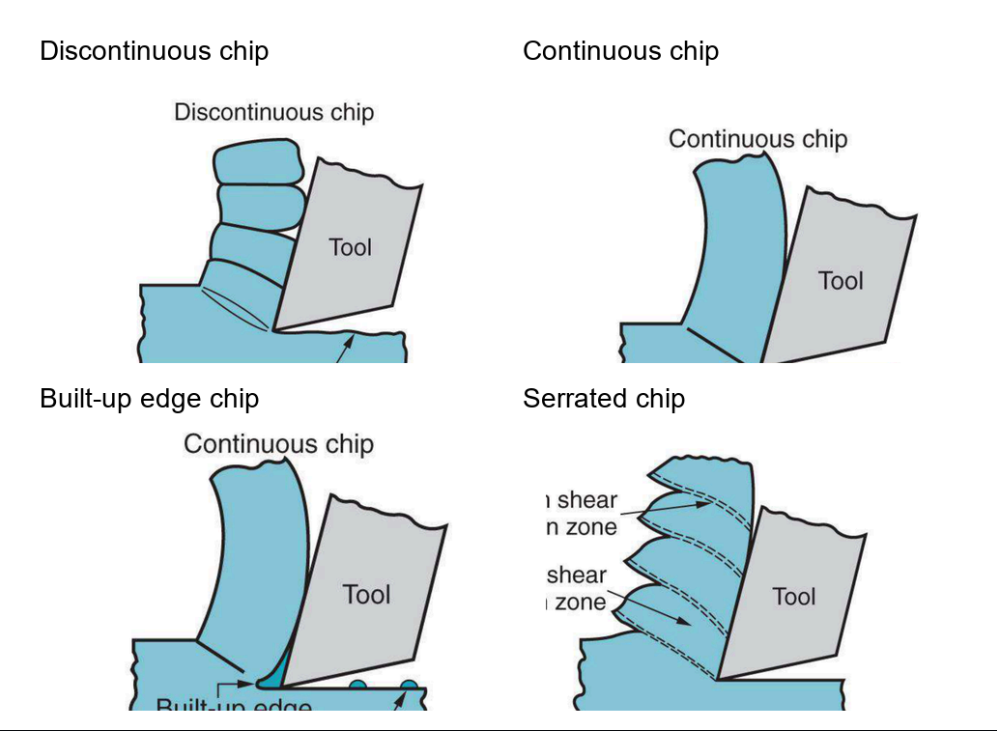


Fig: Types of chips in machining: discontinuous, continuous, built-up edge and serrated

CHIP TYPE	CONDITIONS	MATERIAL
Discontinuous Chips	Low cutting speed, large feed and depth of cut, high tool-chip friction	Brittle work materials (cast iron, brass)
Continuous Chips	High cutting speed, small feeds and depths, sharp cutting edge, low friction	Ductile work materials (mild steel, aluminium)
Continuous Chips with Built-Up Edge (BUE)	Low to medium cutting speeds; tool-chip friction causes chip to adhere to rake face; BUE forms and breaks cyclically	Ductile materials
Serrated (Segmented) Chips	Semi-continuous, saw-tooth appearance; alternating high/low shear strain; high cutting speeds	Difficult-to-machine metals (titanium, stainless steel)

**Built-Up Edge (BUE):** A portion of chip material that adheres to the tool rake face. Cyclically forms and breaks. Causes poor surface finish, tool wear, and dimensional inaccuracy. Avoided by increasing cutting speed or using lubricant.



PREVIOUS YEAR QUESTIONS — UNIT 12 (MACHINING PRINCIPLES)

ETE NOV 20252 marks · CO3

QA3. List any two angles of a single point cutting tool.

Any two from: (1) Back Rake Angle, (2) Side Rake Angle, (3) End Relief/Clearance Angle, (4) Side Relief Angle, (5) End Cutting Edge Angle, (6) Side Cutting Edge Angle. (See Unit 13 for full geometry)

ETE NOV 20256 marks · CO3

QB4. Discuss any four lathe operations and tools used for these operations.

See Unit 14 for full answer.



EXPECTED/SAMPLE QUESTIONS — UNIT 12

 LIKELY

1. What is machining? How does it differ from forming processes? (2 marks)

 WARM

2. Describe the four types of chips produced in metal cutting. (4 marks)

 WARM

3. What is Built-Up Edge (BUE)? How can it be avoided? (3 marks)

 HOT

4. Differentiate between orthogonal and oblique machining. (4 marks)

 LIKELY

5. What is cutting speed and feed speed? What surfaces do they generate? (2 marks)

MATERIAL	PROPERTIES	APPLICATIONS
Plain Carbon Steels	0.6–1.5% carbon; good toughness and abrasion resistance	Hand taps, threading dies, saws (low-speed cutting)
High Speed Steel (HSS)	High carbon + W, Mo, Cr, V, Co; operates up to 600°C; two types: T-type and M-type	Drills, milling cutters, broaches, general-purpose tools
Cast Cobalt Alloys	W + Cr + C + Cobalt alloy; red hardness + wear resistance + toughness	Intermediate applications between HSS and carbides
Cemented Carbides	Carbides of W, Ti, Ta with cobalt binder (powder metallurgy); very hard, 3-4× faster than HSS	High-speed cutting; insert tools on CNC machines
Ceramics	Al ₂ O ₃ , SiC, B ₄ C, TiC, TiB; high speed, long tool life, no coolant needed	High rigidity machines; no interrupted cutting
Cubic Boron Nitride (CBN)	Very high hardness; second hardest after diamond	Hard steel, superalloys
Polycrystalline Diamond (PCD)	Most abrasion resistant; highest hardness; produces mirror finish	Non-ferrous metals, composites, grinding wheels, TV screens

Properties Required in Cutting Tool Materials

- **Red (Hot) Hardness:** Ability to retain hardness at high cutting temperatures
- **Wear Resistance:** Retains shape and cutting efficiency over time
- **Toughness:** Resists shock/impact loads — important for interrupted cuts

| GEOMETRY OF SINGLE POINT CUTTING TOOL

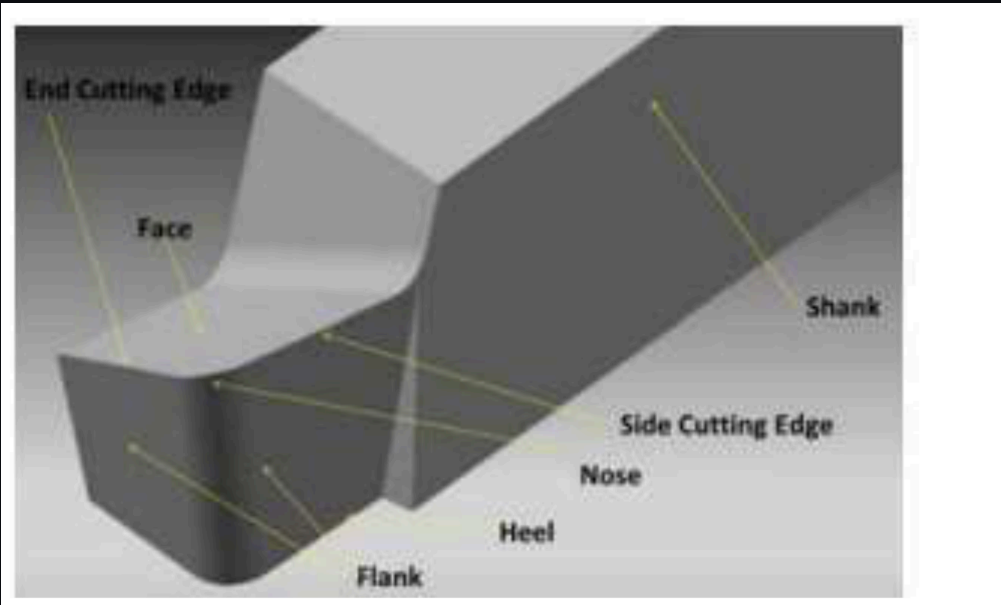


Fig: Single point cutting tool nomenclature showing face, flank, shank, nose and cutting edges

Tool geometry comprises: nose, rake face, flank, heel, and shank.

Tool Angles (6 angles + nose radius = 7 parameters)

#	ANGLE	DESCRIPTION
1	Back Rake Angle	Angle between face of tool and line perpendicular to base, measured on perpendicular plane through side cutting edge. Controls chip flow direction.
2	Side Rake Angle	Angle between face of tool and line parallel to base, measured on plane perpendicular to base and side edge. Affects cutting force and chip thickness.
3	End Relief (Clearance) Angle	Angle between end flank and line perpendicular to base of tool. Prevents rubbing of tool on machined surface.
4	Side Relief Angle	Angle between side flank (below side cutting edge) and line perpendicular to base. Reduces friction on machined surface.
5	End Cutting Edge Angle	Angle between end cutting edge and line perpendicular to shank. Affects surface finish.
6	Side Cutting Edge Angle	Angle between side cutting edge and side of tool shank. Affects chip formation and tool forces.
7	Nose Radius	Radius at tip of tool. Larger radius = better surface finish; smaller = less vibration.

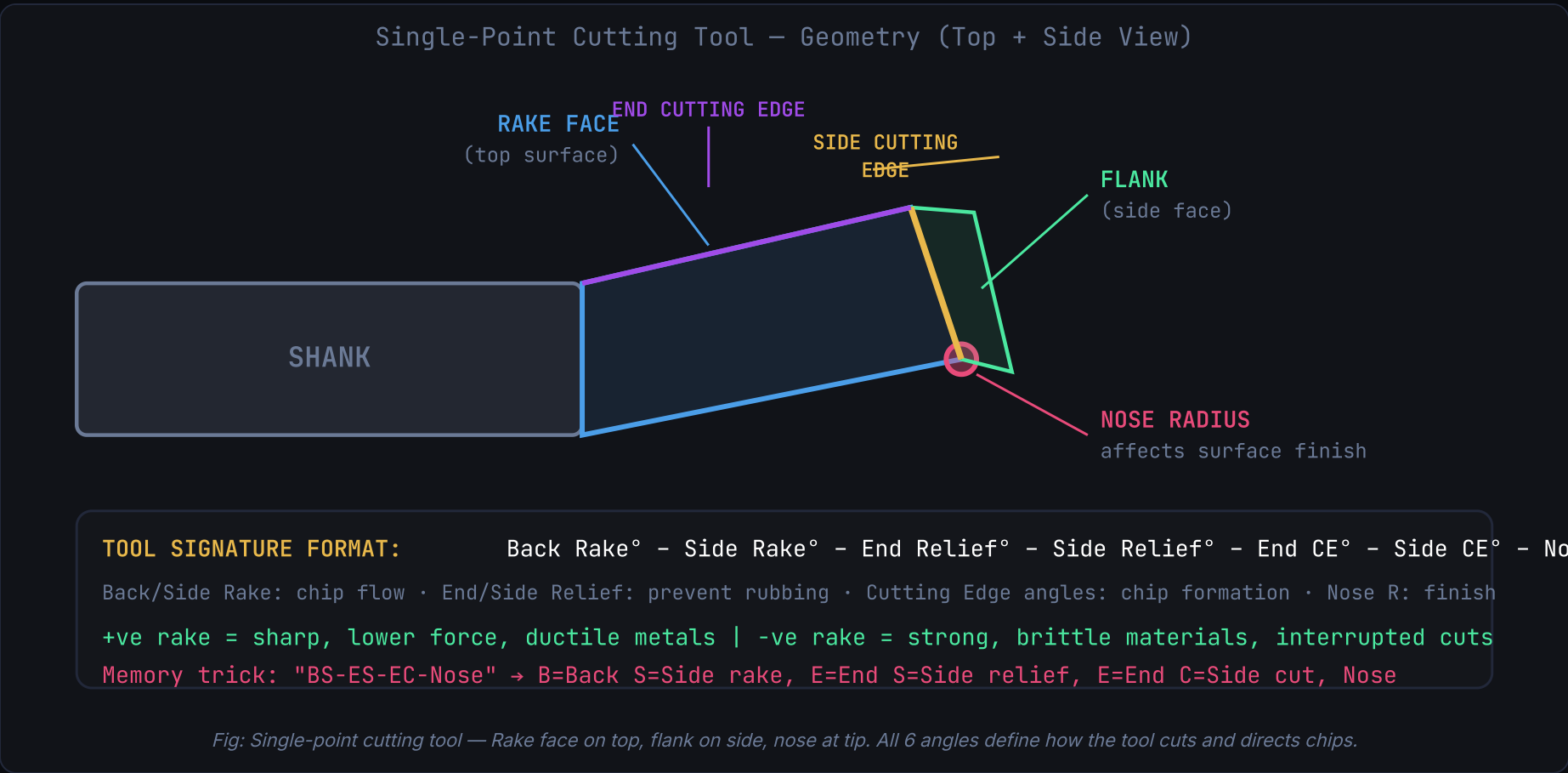
Tool Signature

Tool signature is a special sequence specifying tool geometry:

Format: Back rake – Side rake – End relief – Side relief – End cutting edge – Side cutting edge – Nose radius

Example: 0 - 7 - 6 - 8 - 15 - 16 - 0.8

This means: 0° back rake, 7° side rake, 6° end relief, 8° side relief, 15° end cutting edge, 16° side cutting edge, 0.8 mm nose radius.



I MACHINABILITY

The **ease with which a given workpiece material can be machined with a specified cutting tool.**

Good machinability = low tool wear + good surface finish + low cutting forces + short chip formation.

I TOOL LIFE — TAYLOR'S EQUATION

$$V \times T^n = C$$

Where: V = cutting speed, T = tool life (minutes), n = exponent for conditions, C = Taylor's constant

PREVIOUS YEAR QUESTIONS — UNIT 13 (CUTTING TOOLS)

ETE NOV 2025

2 marks · CO3

QA3. List any two angles of a single point cutting tool.

Any two from the 6 angles: Back Rake Angle and Side Rake Angle (most commonly expected). Or: End Relief Angle, Side Relief Angle, End Cutting Edge Angle, Side Cutting Edge Angle.

RE-SESSIONAL APR 2026

2 marks · CO2

Q2. Name any two cutting tool materials.

High Speed Steel (HSS) and Cemented Carbide (Tungsten Carbide).

EXPECTED/SAMPLE QUESTIONS — UNIT 13

HOT

1. What is tool signature? Write and explain the tool signature 8-10-6-6-8-15-1. (4 marks)

WARM

2. Compare HSS and Cemented Carbide as cutting tool materials. (4 marks)

WARM

3. What is machinability? What factors affect it? (4 marks)

LIKELY

4. What is red hardness? Why is it important for cutting tools? (2 marks)

WARM

5. State Taylor's Tool Life equation and explain each term. (3 marks)

| WHAT IS A LATHE?

A lathe is a machine tool that **removes metal from a workpiece by rotating the workpiece about its axis** while a fixed cutting tool is fed into it. *Principle: Rotating workpiece + fixed cutting tool.*

| MAIN PARTS OF A LATHE

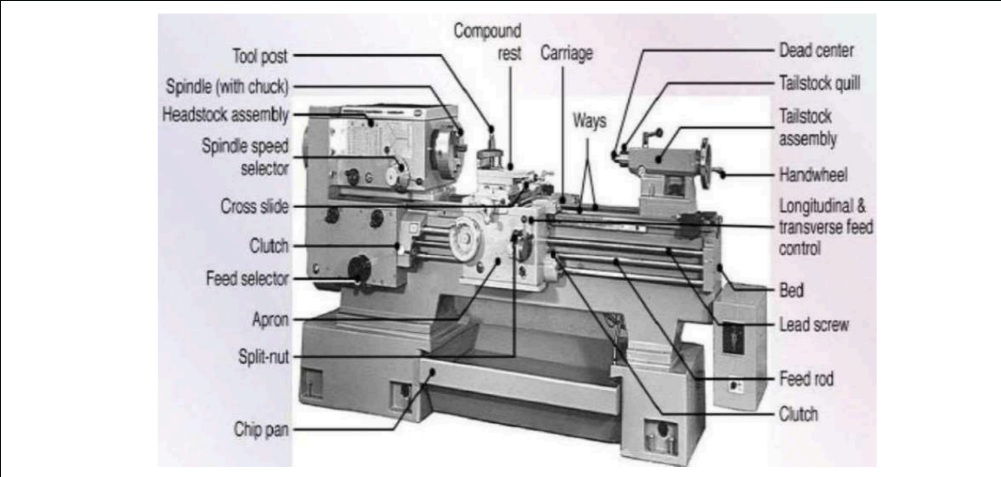


Fig: Lathe machine showing headstock, tailstock, carriage, bed, tool post and lead screw

PART	FUNCTION
Headstock	Houses the spindle and speed control mechanism. Mounted on left end of lathe bed. Drives the workpiece.
Spindle	Hollow shaft in headstock that rotates the workpiece. Work or chuck attached to it.
Chuck	Device mounted on spindle to hold and rotate the workpiece. Types: 3-jaw (self-centering) and 4-jaw (independent).
Bed	Main foundation of lathe — heavy casting providing rigidity and supporting all components. Has guide ways (ways).
Tailstock	Mounted on right end of bed. Holds long work (with center) or drill/reamer in its quill. Can slide along bed.
Carriage	Moves along bed ways — carries the cutting tool. Components: Saddle, Cross-slide, Compound rest, Tool post.
Tool Post	Holds the cutting tool securely. Mounted on compound rest.
Lead Screw	Long threaded rod driven by headstock. Used for threading operations and auto-feed. Essential for screw cutting.
Feed Rod	Drives the carriage automatically for turning operations (not threading).
Apron	Front face of carriage — houses mechanisms for engaging feed and lead screw.

| LATHE OPERATIONS

OPERATION	DESCRIPTION	TOOL FEED DIRECTION
Turning	Produces cylindrical or conical external surfaces by reducing workpiece diameter	Parallel to axis of rotation
Facing	Produces flat surface at end of workpiece (the face). Reduces workpiece length.	Perpendicular (90°) to axis
Taper Turning	Produces tapered (cone-type) surface on workpiece	At an angle to axis
Drilling	Produces cylindrical hole by fixing drill bit in tailstock; workpiece rotates	Along axis (tailstock)
Boring	Enlarges an existing hole made by drilling; improves hole diameter and surface finish	Parallel to axis (internal)
Threading (Screw Cutting)	Produces external or internal threads. Lead screw relationship determines pitch.	Parallel to axis (using lead screw)
Knurling	Embosses diamond-shaped pattern on surface to provide gripping surface. No material removed.	Rolling action against surface
Chamfering	Beveling the extreme end/edge of workpiece to remove burrs and sharp edges	At 45° to axis
Grooving (Necking)	Reduces diameter at a desired portion. Creates groove/slot on cylindrical surface.	Perpendicular (in-feed)
Parting	Cuts off a portion of workpiece. Separates finished part from bar stock.	Perpendicular (radially inward)
Reaming	Finishes and sizes a drilled hole for accuracy and smooth finish using a reamer	Along axis

QD1. Give any 5 parts of Lathe machine and give their functions in detail.

Any 5 from the table above with detailed descriptions. For 10-mark answer, write 4–5 lines each:

- **Headstock:** Fixed at left end of lathe bed. Houses spindle, gear box, and speed control. Drives the workpiece at required RPM via the spindle.
- **Bed:** Main foundation — heavy grey iron casting. Provides rigidity, houses guide ways for carriage and tailstock movement. All other parts mounted on bed.
- **Tailstock:** Supports long workpieces using dead/live centre. Can hold drilling tools (drill, reamer, tap) in its quill. Slides along bed ways; locked in position.
- **Carriage:** Moves along guide ways. Holds cutting tool through tool post. Carries saddle, cross-slide, compound rest, and apron.
- **Lead Screw:** Long precision screw driven by headstock gears. Used specifically for thread cutting operations. Gear ratio between spindle and lead screw determines thread pitch.
- **Tool Post:** Holds cutting tool at correct height and angle. Mounted on compound rest. Can be swivelled for angular cuts.
- **Chuck:** Workholding device on spindle. 3-jaw self-centering for round work; 4-jaw independent for square/irregular work.

Q5. Describe operations of milling machine. (Note: Milling is a lathe-related multi-point machining — see Unit 15)

See Unit 15 for milling operations.

Q1. List any two Lathe machine parts.

Headstock and Tailstock. (Or: Carriage and Bed; Chuck and Lead Screw — any two with brief function)

QB4. Discuss any four lathe operations and tools used for these operations.

- **Turning:** Tool: Single-point turning tool. Workpiece rotates, tool fed parallel to axis. Reduces diameter to produce cylindrical surface.
- **Facing:** Tool: Facing tool (single-point, fed at 90° to axis). Produces flat surface at end of workpiece.
- **Drilling:** Tool: Twist drill mounted in tailstock chuck. Produces cylindrical hole as workpiece rotates.
- **Threading:** Tool: Threading/screw cutting tool (V-form tool). Lead screw ensures synchronized motion for correct pitch.
- **Knurling:** Tool: Knurling tool (hardened serrated wheel). Embosses diamond pattern for grip (e.g., knobs, handles).
- **Boring:** Tool: Boring bar with single-point tool. Enlarges an existing drilled hole for accuracy.

💡 EXPECTED/SAMPLE QUESTIONS — UNIT 14

💧 LIKELY

1. Describe the working principle of a lathe machine. (2 marks)

⚡ WARM

2. What is the difference between turning and facing operations? (2 marks)

⚡ WARM

3. Explain knurling operation. What is its purpose? What tool is used? (3 marks)

🔥 HOT

4. What is the role of lead screw in a lathe? (2 marks)

⚡ WARM

5. Differentiate between boring and drilling operations on a lathe. (4 marks)

Milling & Drilling

I MILLING

Milling is a **multi-point machining process** where material is removed by a rotating multi-toothed cutter (milling cutter). The workpiece is fed against the rotating cutter.

Types of Milling Operations

OPERATION	DESCRIPTION
Face Milling	Cutter axis is perpendicular to workpiece surface. Produces flat surfaces on top of workpiece.
Peripheral (Plain) Milling	Cutter axis is parallel to workpiece surface. Cutting done by teeth on periphery.

OPERATION	DESCRIPTION
Slot Milling	Narrow groove/slot machined into workpiece.
End Milling	End mill cutter used for slots, pockets, contours.
Gang Milling	Multiple cutters on same arbor cutting simultaneously.
Profile Milling	Milling curved or irregular profiles.

Types of Milling (by Cutter Rotation vs Feed)

- **Up (Conventional) Milling:** Cutter rotation opposes direction of feed. Chip thickness increases from zero to maximum. Safer (backlash not a problem).
- **Down (Climb) Milling:** Cutter rotation in same direction as feed. Chip thickness decreases from maximum to zero. Better surface finish, but backlash can be a problem.

I DRILLING

Drilling is a **multi-point machining operation** that produces cylindrical holes in a workpiece using a rotating drill bit.

Parts of a Drilling Machine

PART	FUNCTION
Base	Foundation of drilling machine; supports all other components
Column	Vertical support connecting base to head; allows table height adjustment
Table	Work-holding surface; can be raised/lowered along column; has T-slots for clamps
Head (Spindle Housing)	Houses spindle, motor, and feed mechanism at top of column
Spindle	Rotating shaft holding drill bit through chuck or morse taper
Chuck	Holds the drill bit; keyless or keyed type
Feed Handle	Moves spindle downward for drilling; usually 3-arm handle
Depth Stop	Limits depth of drilling

Drilling Operations

- **Drilling:** Producing cylindrical holes using twist drill
- **Reaming:** Finishing and sizing a drilled hole — uses multi-fluted reamer
- **Boring:** Enlarging hole with single-point tool for accuracy
- **Counterboring:** Enlarging top portion of hole — creates stepped/shouldered hole for bolt heads
- **Countersinking:** Making conical widening at top of hole for countersunk screws
- **Tapping:** Cutting internal threads in a drilled hole using a tap
- **Spot Facing:** Smoothing a circular area on rough surface around a hole

 PREVIOUS YEAR QUESTIONS — UNIT 15 (MILLING & DRILLING)

RE-SESSIONAL APR 2026 4 marks · CO1

Q5. Describe operations of milling machine.

Main milling operations:

- **Face Milling:** Cutter axis perpendicular to workpiece; produces flat surfaces. Most common milling operation.
- **Peripheral (Plain) Milling:** Cutter axis parallel to workpiece; peripheral teeth do cutting; produces flat surface along workpiece length.
- **Slot Milling:** Side milling cutter used to machine narrow slots.
- **End Milling:** End mill cutter used for vertical and horizontal surfaces, slots, and pockets. Like a drill that also cuts sideways.
- **Gang Milling:** Multiple cutters mounted on same arbor; multiple surfaces machined simultaneously.
- **Profile Milling:** Produces curved, irregular profiles using form milling cutters.

RE-SESSIONAL NOV 2025 4 marks · CO3

Q6. Describe parts of drilling machine.

See parts table above. For exam, name and describe: Base, Column, Table, Spindle Housing/Head, Spindle, Chuck, Feed Handle.

💡 EXPECTED/SAMPLE QUESTIONS — UNIT 15

- 💧 LIKELY

1. What is milling? How does it differ from turning? (2 marks)
- ⚡ WARM

2. Differentiate between up milling and down milling. (4 marks)
- 🔥 HOT

3. List and briefly explain four drilling operations. (4 marks)
- 💧 LIKELY

4. What is counterboring? When is it used? (2 marks)
- ⚡ WARM

5. Explain face milling with a neat sketch. (4 marks)

UNIT 16

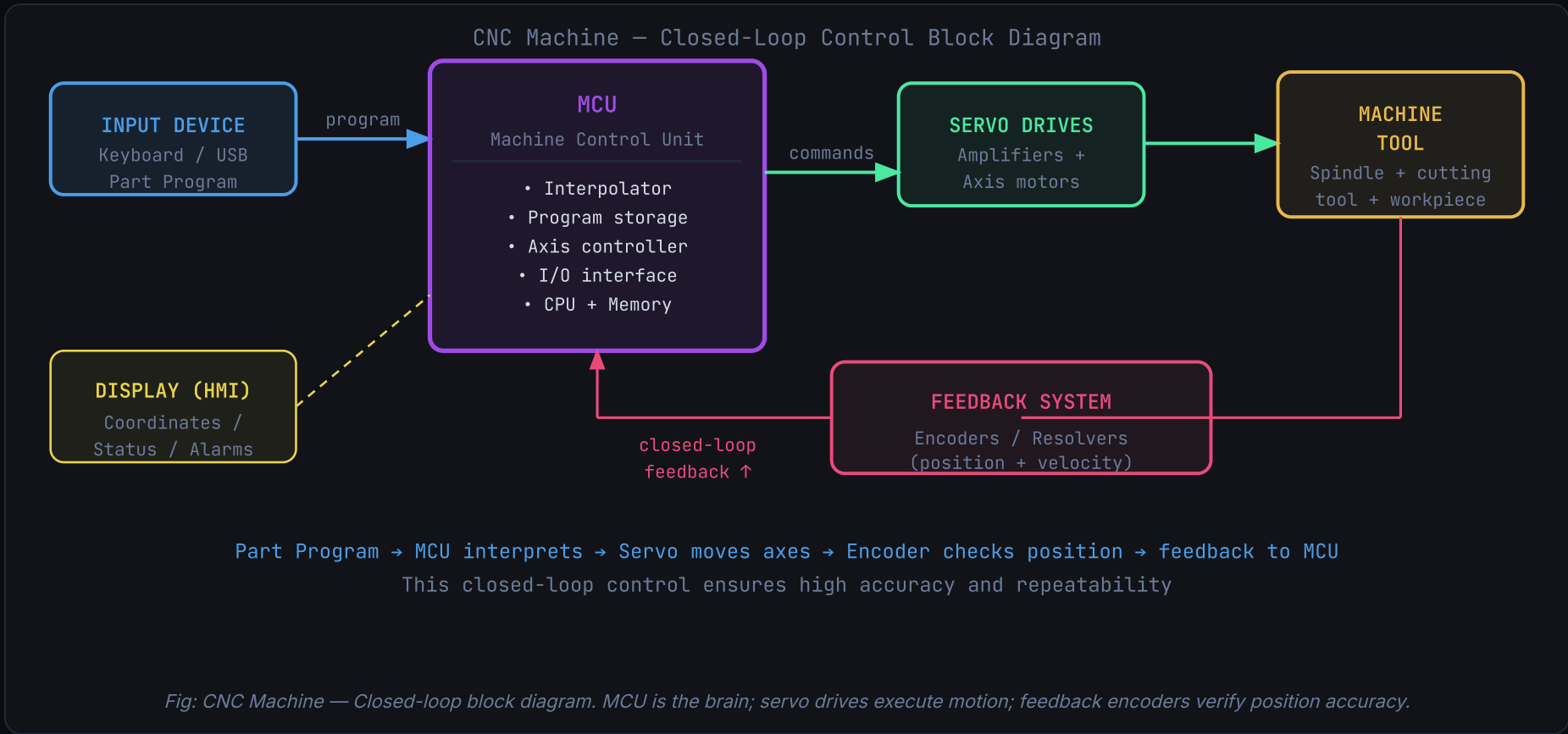
CNC: Computerized Numerical Control

I INTRODUCTION TO CNC

CNC (Computerized Numerical Control) is a method of controlling machine tool operations by means of a computer program containing coded instructions (G and M codes). The machine automatically follows the programmed path.

I COMPONENTS OF A CNC MACHINE

COMPONENT	FUNCTION
Machine Control Unit (MCU)	The "brain" — reads and interprets part program, generates motion commands for each axis
Machine Tool	The physical cutting machine (lathe, milling, drill) — performs actual machining
Servo Drive System	Converts electrical signals from MCU into mechanical motion of axes
Feedback System	Position and velocity feedback (encoders) to verify actual vs commanded position — closed-loop control
Input Device	Keyboard, tape reader, USB — for entering part program
Display Unit (HMI)	Shows machine status, coordinates, program execution



I ADVANTAGES OF CNC MACHINES

- High accuracy and repeatability
- Complex profiles machined easily
- Reduced human error
- Consistent quality across batches
- Less skilled operator required for machining (skilled for programming)
- Faster setup and changeover
- Suitable for small to medium batch production

I G-CODES (PREPARATORY FUNCTIONS)

G-CODE	FUNCTION
G00	Rapid positioning (move at maximum speed to next location)

G-CODE	FUNCTION
G01	Linear interpolation (move in straight line at programmed feed rate)
G02	Circular interpolation — clockwise (CW)
G03	Circular interpolation — counterclockwise (CCW)
G04	Dwell (pause for specified time)
G17	XY plane selection
G20	Inch units (imperial)
G21	Metric units (mm)
G28	Return to machine home/reference point
G40	Cancel cutter diameter compensation
G41	Cutter compensation — left
G42	Cutter compensation — right
G43	Tool length compensation — positive offset
G54–G59	Work coordinate system selection
G80	Cancel canned cycle
G81	Drilling canned cycle
G90	Absolute programming mode
G91	Incremental programming mode

I M-CODES (MISCELLANEOUS FUNCTIONS)

M-CODE	FUNCTION
M00	Program stop (unconditional stop)
M01	Optional stop
M02	End of program
M03	Spindle ON — clockwise (CW)
M04	Spindle ON — counterclockwise (CCW)
M05	Spindle stop (OFF)
M06	Automatic tool change (ATC)
M08	Coolant ON
M09	Coolant OFF
M30	End of program + rewind (reset to beginning)

 **Exam tip:** "G01 and G03 codes in CNC programming stand for?" → G01 = Linear Interpolation (straight line at feed rate); G03 = Circular Interpolation CCW. This exact question appears in ETE Nov 2025 (QA4). Know G00, G01, G02, G03, G90, G91, M03, M05, M06, M30 thoroughly.

I CNC PART PROGRAMMING FORMAT

A typical CNC program block format:

```
N [Block Number] G [Preparatory] X [X coord] Y [Y coord] Z [Z coord] F [Feed] S [Speed] T [Tool] M [Misc] Example: N001 G21
→ Metric units N002 G90 G54 → Absolute mode, Work coordinate system N003 M06 T01 → Tool change to Tool 1 N004 M03 S1000 →
Spindle ON CW at 1000 RPM N005 G00 X0 Y0 Z5 → Rapid to X0,Y0, Z5mm above N006 G01 Z-1.5 F100 → Linear plunge to Z-1.5mm at
100mm/min N007 G01 X60 F150 → Cut along X to 60mm N008 G01 Y30 → Cut along Y to 30mm N009 G00 Z5 → Rapid retract N010 M05 →
Spindle OFF N011 M30 → End of program
```

I ETE SAMPLE PROGRAM — SLOT MACHINING

Workpiece: 60mm × 30mm × 6mm. Slot: 1.5mm deep. Cutter diameter: 8mm.

CNC Part Program – Slot MachiningG & M codes

```
%
                                (Program start)
O0001                          (Program number)
N01 G21 G17 G40                (Metric, XY plane, cancel compensation)
N02 G90 G54                    (Absolute mode, work coordinate)
N03 T01 M06                    (Select tool 1 – 8mm end mill, change)
N04 S1500 M03                  (Spindle 1500 RPM, CW)
N05 G00 X0 Y0 Z10.0            (Rapid to start position above work)
N06 G00 X5 Y15                 (Rapid to slot start position)
N07 G01 Z-1.5 F80              (Plunge feed to 1.5mm depth at 80mm/min)
N08 G01 X55 F150               (Cut slot along X direction, 60-5=55mm)
N09 G00 Z10.0                  (Rapid retract up)
N10 G00 X0 Y0                  (Return to origin)
N11 M05                        (Spindle stop)
N12 M30                        (End of program, rewind)
%
```

PREVIOUS YEAR QUESTIONS — UNIT 16 (CNC)

ETE NOV 2025

2 marks · CO4

QA4. G01 and G03 codes in CNC programming stand for?

G01: Linear Interpolation — commands the machine to move in a straight line at the programmed feed rate (F value). Used for straight cutting moves.

G03: Circular Interpolation Counter-Clockwise (CCW) — commands the machine to move in an arc in the counter-clockwise direction at programmed feed rate.

RE-SESSIONAL NOV 2025

4 marks · CO4

Q7. What are the applications of CNC machines?

Applications of CNC machines:

- Aerospace Industry:** Complex turbine blades, structural components, aircraft frames
- Automotive Industry:** Engine blocks, cylinder heads, transmission components, molds for body panels
- Medical Devices:** Implants (hip joints, dental implants), surgical instruments requiring high precision
- Electronics:** PCB drilling, enclosure machining, heat sinks
- Die & Mould Making:** Injection moulding dies, stamping dies with complex profiles
- Prototype Manufacturing:** Rapid production of one-off precision parts
- Defence:** Weapons components, navigation systems, precision parts

RE-SESSIONAL APR 2026

2 marks · CO3

Q3. List any two miscellaneous functions used for CNC programming.

- M03:** Spindle ON — Clockwise (CW). Starts the spindle rotation in clockwise direction at programmed speed.
- M05:** Spindle Stop. Stops the spindle rotation.
- (Also acceptable: M06 — Tool Change; M08 — Coolant ON; M30 — End of Program + Rewind)

RE-SESSIONAL APR 2026

9 marks · CO2

Q8. Prepare sketch of components of CNC machine and describe its various components.

CNC Machine Components:

 See the **CNC closed-loop block diagram SVG** above in the Unit 16 components section — use that diagram for the sketch. Label all 6 components clearly.

Components described:

- Input Device:** Keyboard, USB, tape reader — for entering part programs
- Machine Control Unit (MCU):** Computer that reads program, performs calculations, generates motion commands. Has CPU, memory, I/O interface.
- Servo Drive System:** Amplifiers and motors that physically move each axis according to MCU commands
- Feedback System:** Rotary encoders or linear scales feed back actual position to MCU — enables closed-loop control
- Machine Tool:** Physical cutting machine (milling, turning, drilling) with spindle, axes, workholding

- **Display (HMI):** Shows program execution, coordinates, alarms, machine status

ETE NOV 2025 10 marks · CO4

QD2. Describe basic components of CNC machine. Prepare a part program using G-codes and M-codes for machining any 1.5 mm deep slot in a work piece of 60 mm × 30 mm × 6 mm. Cutter diameter = 8 mm, spindle speed = ?

Components: (Same as Q8 answer above)

Part Program: See CNC Part Program example in the notes section above for complete slot machining program.

💡 EXPECTED/SAMPLE QUESTIONS — UNIT 16

⚡ WARM 1. What is CNC? What are its advantages over conventional machining? (4 marks)

⚡ WARM 2. What is the difference between G90 and G91 in CNC? (2 marks)

🔥 HOT 3. Write a CNC program to drill a hole at (20,15) to depth 10mm on a 50×50mm workpiece. (6 marks)

💧 LIKELY 4. What is the purpose of M06 code? (1 mark)

🔥 HOT 5. Explain the difference between G00 and G01. When would you use each? (2 marks)

UNIT 17

Advanced Machining: USM & AJM

WHY ADVANCED/NON-CONVENTIONAL MACHINING?

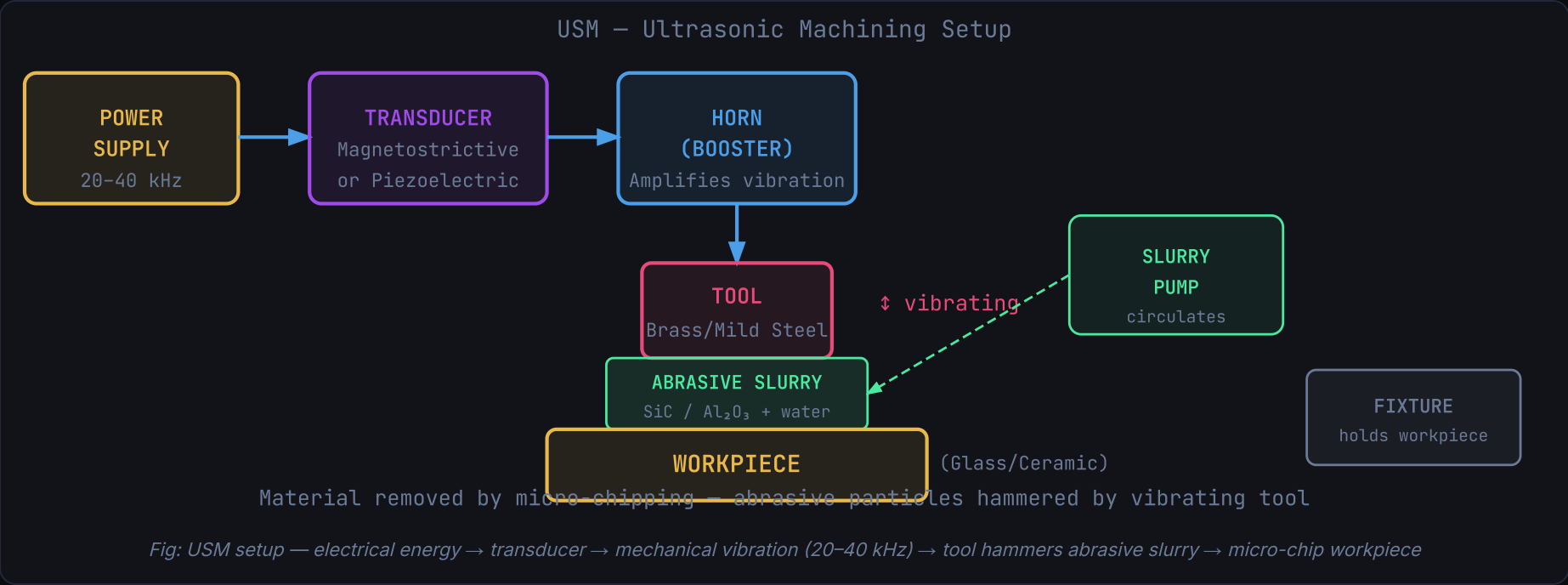
CONVENTIONAL MACHINING	NON-CONVENTIONAL (ADVANCED) MACHINING
Uses direct mechanical contact of tool	Uses electrical, chemical, thermal, or abrasive energy
Limited to machinable hardness levels	Can machine any material regardless of hardness
Hard to machine brittle, fragile, thin parts	Ideal for brittle, hard, thin, heat-sensitive materials
Tool wear is significant issue	Often no physical tool-workpiece contact
Good for ductile metals	Good for ceramics, glass, composites, superalloys

ULTRASONIC MACHINING (USM)

USM is a non-traditional machining process in which material is removed from a **hard, brittle workpiece** surface by action of abrasive particles (in a slurry) driven against the workpiece by high-frequency (20–40 kHz) vibrations of a tool.

Components of USM Machine

1. **Power Supply:** Provides high-frequency electrical energy (≈20–40 kHz)
2. **Transducer:** Converts electrical energy to mechanical vibrations — magnetostrictive (nickel alloy) or piezoelectric (quartz, barium titanate)
3. **Horn (Booster):** Amplifies and transmits ultrasonic vibrations to tool
4. **Tool:** Made of ductile material (brass, mild steel, stainless steel) — shaped as required cavity/hole. Must have high wear resistance and fatigue strength.
5. **Abrasive Slurry System:** Mixture of abrasive particles (SiC, Al₂O₃, B₄C) and water; pumped to machining zone
6. **Workpiece:** Usually hard, brittle, non-conductive materials (glass, ceramics, quartz, carbides)
7. **Fixture & Tank:** Workpiece held in tank filled with slurry for stability



USM Advantages & Disadvantages

ADVANTAGES	DISADVANTAGES
Machines hard, brittle, non-conductive materials (glass, ceramics, carbides)	Low MRR — slow process, not for bulk removal
No heat-affected zone — no thermal damage	Tool wear is significant
Stress-free, burr-free, crack-free surfaces	Limited to small and shallow holes
Intricate shapes and fine holes with good accuracy	Cannot machine ductile materials effectively
No chemical or electrical requirements (unlike ECM/EDM)	High initial setup cost

USM Applications

- Machining ceramics, glass, quartz, silicon, carbides
- Precise hole drilling in hard materials (dies, jewel bearings)
- Shaping of intricate die cavities
- Surface texturing, engraving, frosting of glass
- Micro-drilling of semiconductor wafers
- Aerospace: machining hard composites and heat-resistant materials

| ABRASIVE JET MACHINING (AJM)

AJM removes material from a workpiece by abrasive particles entrained in a **high-velocity gas stream**.

Components of AJM

- Gas supply:** Nitrogen, CO₂, or filtered air at 2–8 kg/cm² (690–830 kPa). Never oxygen (chemical reaction hazard).
- Gas filter and water separator**
- Vibrating chamber:** Mixes gas and abrasive particles
- Nozzle:** Made of tungsten carbide (12–30 hrs life) or sapphire (300 hrs life). Round or rectangular holes.
- Dust hood/collection cabinet:** Collects abrasive waste

Abrasives Used in AJM

- Aluminium oxide (Al₂O₃) and Silicon carbide (SiC) — heavy cleaning, cutting, deburring
- Magnesium carbonate — light cleaning, etching
- Sodium bicarbonate — fine cleaning, soft material cutting
- Optimal grain size: 10–50 μm with sharp edges

AJM Advantages & Disadvantages

ADVANTAGES	DISADVANTAGES
Best for brittle, heat-sensitive materials (glass, quartz, ceramics)	Slow material removal rate
No tool changes required	Low accuracy (±0.1 mm); stray cutting unavoidable
Intricate parts with sharp corners possible	Tapering effect may occur when drilling
Workpiece not hardened; no thermal damage	Abrasive may get embedded in workpiece surface
Not reactive with any workpiece material	Soft materials cannot be machined
Can machine thin materials	Silica dust is health hazard; need dust collection

AJM Applications

- Drilling holes, cutting slots, deburring, polishing, radiusing
- Deburring precision parts (hydraulic valves, aircraft fuel systems, medical devices)
- Machining intricate shapes in brittle/thin materials
- Insulation stripping and wire cleaning
- Micro-deburring of hypodermic needles
- Frosting glass, trimming circuit boards



PREVIOUS YEAR QUESTIONS — UNIT 17 (USM & AJM)

ETE NOV 2025

2 marks · CO5

QA5. How ultrasonic machining process is different from abrasive jet machining?

ASPECT	USM	AJM
Medium	Abrasive slurry (liquid + abrasives)	Gas stream carrying abrasive particles
Tool	Vibrating tool (20–40 kHz)	No physical tool — gas nozzle only
Material removal	Micro-chipping by tool impact on abrasives	Erosion by high-velocity abrasive particles
Accuracy	Better accuracy; intricate shapes	Lower accuracy (±0.1 mm); stray cutting
Application	Deep holes, dies, moulds in hard brittle materials	Surface cleaning, deburring, thin materials

ETE NOV 2025

6 marks · CO5

QB3. Prepare a chart of various unconventional machining processes based on type of energy used for material removal, features and applications.

PROCESS	ENERGY TYPE	MATERIAL REMOVAL	BEST FOR
USM	Mechanical (Ultrasonic vibration)	Micro-chipping by abrasive particles	Hard, brittle non-conductive materials
AJM	Mechanical (Kinetic energy of gas+abrasive)	Erosion by abrasive impact	Brittle materials, deburring, cleaning
ECM	Electrochemical (anodic dissolution)	Ion removal in electrolyte	Hard conductive metals; no tool wear
EDM	Thermal (Electrical discharge/spark)	Spark erosion	Hard conductive metals, dies, moulds
LBM	Radiation (Laser beam)	Melting/vaporization by laser	Thin materials, precision cutting, any material
EBM	Radiation (Electron beam)	Melting/vaporization	Very precise micro-holes; in vacuum



EXPECTED/SAMPLE QUESTIONS — UNIT 17



HOT 1. Explain the principle and components of USM with a neat sketch. (6 marks)



WARM 2. What type of transducer is used in USM? Explain piezoelectric and magnetostrictive types. (4 marks)



WARM 3. What abrasives are used in AJM? Why is oxygen not used as carrier gas? (3 marks)



LIKELY 4. What is stand-off distance (NTD) in AJM? How does it affect VRR? (3 marks)



WARM 5. Why can USM not be used on ductile materials effectively? (2 marks)

ECM removes material by **controlled electrochemical dissolution (anodic dissolution)** — the workpiece is the anode (+) and the tool is the cathode (–) in an electrolyte.

Principle

Based on Faraday's law of electrolysis: When DC current flows between tool (cathode) and workpiece (anode) through an electrolyte, metal ions from the workpiece dissolve into the electrolyte. The shape of the tool is replicated (mirror image) on the workpiece.

Process Parameters

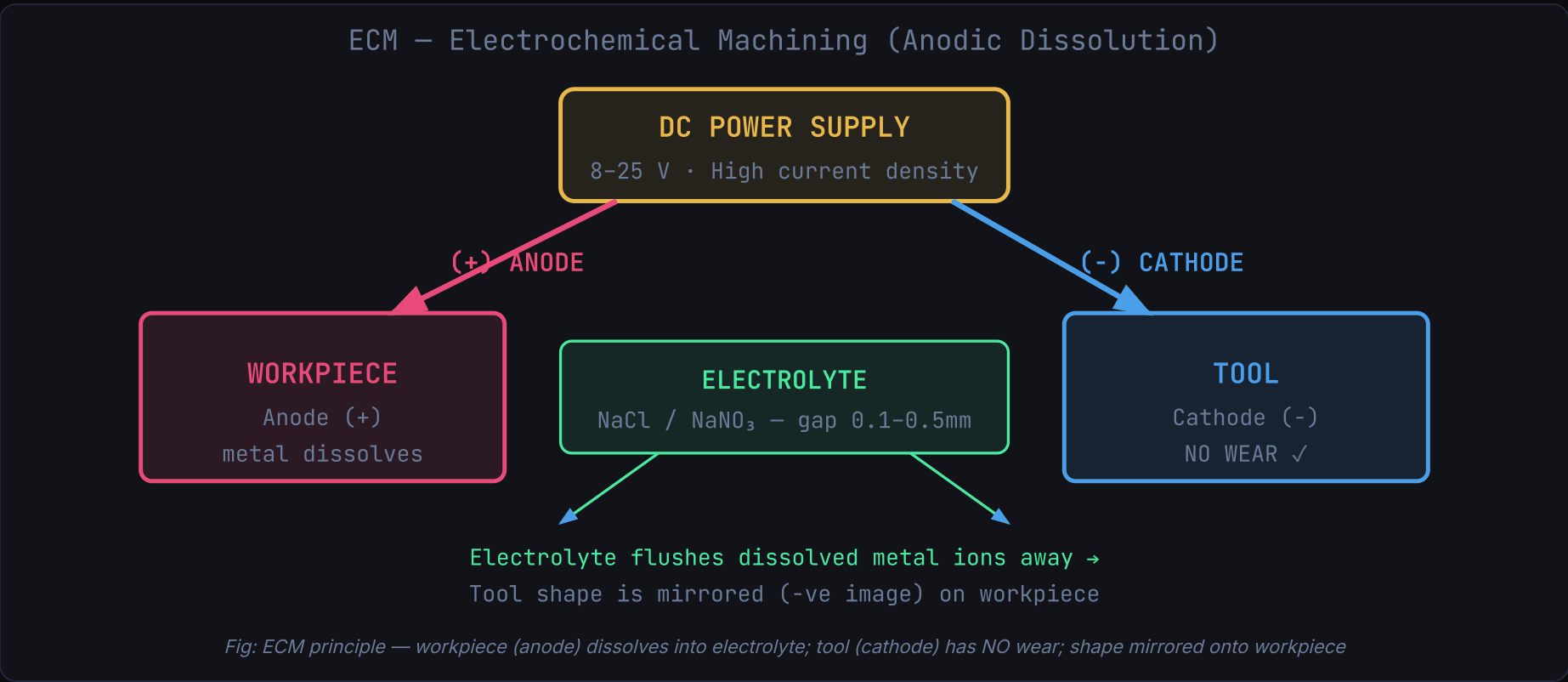
- **Electrolyte:** NaCl (sodium chloride) or NaNO₃ (sodium nitrate) solution — pumped between tool and workpiece
- **Voltage:** 8–25 V DC
- **Current density:** Very high (up to 300 A/cm²)
- **Gap between tool and work:** Very small (0.1–0.5 mm)
- **Tool material:** Copper, brass, stainless steel (does not wear)

ECM Advantages & Disadvantages

ADVANTAGES	DISADVANTAGES
No tool wear (tool is cathode, not consumed)	Only conductive materials can be machined
No mechanical stresses or thermal damage	High initial cost, electrolyte handling required
Very smooth surface finish	Low accuracy due to electrolyte spreading
Complex 3D shapes easily machined	Electrolyte disposal is an environmental concern
High MRR compared to EDM	Not suitable for non-conducting materials

ECM Applications

- Machining hard-to-cut metals (turbine blades, superalloys)
- Die sinking for intricate die shapes
- Drilling deep holes in hard steel
- Deburring of precision parts



| ELECTRIC DISCHARGE MACHINING (EDM)

EDM removes material by **controlled electrical sparks (discharges)** between a tool electrode and workpiece. Both are submerged in a dielectric fluid.

Principle

When voltage applied between tool and workpiece (in dielectric), the gap breaks down and a spark (discharge) occurs. Each spark generates extremely high localized temperature (8000–12000°C), melting and vaporizing a tiny amount of workpiece material. This repeats at high frequency, gradually eroding the workpiece into the shape of the tool.

Process Parameters

- **Dielectric fluid:** Kerosene, deionized water — insulates, flushes debris, cools
- **Voltage:** 50–300 V
- **Discharge current:** 0.5–500 A
- **Pulse frequency:** Thousands of discharges per second
- **Gap between tool and work:** 0.025–0.5 mm (spark gap)
- **Tool material:** Graphite (most common), copper, copper-tungsten

EDM Advantages & Disadvantages

ADVANTAGES	DISADVANTAGES
Can machine any electrically conductive material regardless of hardness	Only conductive materials can be machined
Complex shapes and deep cavities easily produced	Slow material removal rate
No cutting forces — no mechanical distortion	Tool (electrode) wears out
Excellent surface finish possible	HAZ (heat-affected zone) created on workpiece surface
Narrow slots and deep holes possible	High energy consumption

EDM Applications

- Die and mould making (injection moulds, die casting dies)
- Machining of hardened steel — no need to machine before hardening
- Aerospace components (turbine blades)
- Wire EDM for cutting complex 2D profiles
- Medical implants

| ECM VS EDM COMPARISON

FEATURE	ECM	EDM
Principle	Electrochemical dissolution (anodic)	Spark erosion (thermal)
Medium	Electrolyte (NaCl, NaNO ₃)	Dielectric fluid (kerosene, DI water)
Tool wear	No tool wear	Tool (electrode) wears
Thermal damage	No heat — no HAZ	HAZ present; residual stresses
Surface finish	Very smooth	Good (with fine finishing passes)
MRR	Higher	Lower
Material requirement	Must be electrically conductive	Must be electrically conductive



PREVIOUS YEAR QUESTIONS — UNIT 18 (ECM & EDM)

RE-SESSIONAL NOV 2025

8 marks · CO5

Q8. How electrochemical machining is used to obtain desired shape of metal part?

Electrochemical Machining (ECM) Process:

- ECM uses the principle of electrolysis (Faraday's law) for material removal
- **Setup:** Workpiece connected as *anode (+)*; Tool connected as *cathode (-)*; Both submerged in electrolyte (NaCl or NaNO₃)
- **DC current** (8–25 V, very high current density) flows through the electrolyte
- Metal atoms from workpiece (anode) dissolve as positive ions into electrolyte — **anodic dissolution**
- Tool slowly moves toward workpiece maintaining constant small gap (0.1–0.5 mm)
- Shape of tool is replicated as a mirror image on workpiece
- Electrolyte also carries away dissolved material and heat
- **Key advantage:** Tool does NOT wear. No thermal effects.

Applications: Turbine blades, die sinking, deep drilling in superalloys, deburring.

ETE NOV 2025

10 marks · CO5

QC2. Describe Electric Discharge Machining process, its parameters and applications.

EDM Principle: Material removed by controlled electrical sparks between tool electrode (–) and workpiece (+) submerged in dielectric fluid.

Process: High voltage (50–300V) applied → when gap small enough, dielectric breaks down → spark occurs → temperature reaches 8000–12000°C locally → tiny volume of workpiece melts/vaporizes → debris flushed by dielectric → process repeats thousands of times per second → gradually erodes workpiece into tool shape.

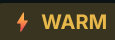
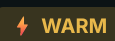
Parameters:

- Dielectric fluid: Kerosene or deionized water
- Voltage: 50–300 V; Current: 0.5–500 A

- Pulse on/off time, Duty cycle
- Spark gap: 0.025–0.5 mm
- Tool material: Graphite (most common), copper

Applications: Die/mould making, hardened steel machining, turbine blades, complex profiles, Wire EDM for 2D contours, medical implants.

💡 EXPECTED/SAMPLE QUESTIONS — UNIT 18

-  **HOT** 1. Explain the principle of ECM. Why doesn't the tool wear in ECM? (4 marks)
-  **WARM** 2. What is dielectric fluid in EDM? What functions does it serve? (3 marks)
-  **HOT** 3. Compare ECM and EDM on any five aspects. (5 marks)
-  **WARM** 4. Why are both ECM and EDM limited to electrically conductive materials? (2 marks)
-  **LIKELY** 5. What is Wire EDM? Where is it used? (2 marks)

UNIT 19

Laser Beam Machining (LBM) & Electron Beam Machining (EBM)

| LASER BEAM MACHINING (LBM)

LBM uses a **highly focused, high-energy laser beam** to melt, vaporize, or chemically change workpiece material. The word LASER = *Light Amplification by Stimulated Emission of Radiation*.

Principle

- Laser generates a coherent, monochromatic, highly collimated beam of light
- Beam is focused by lens to a very small spot (0.01–0.1 mm diameter)
- Extremely high power density at focus point melts and vaporizes material
- Material removal by thermal ablation — no direct tool contact

Types of Lasers Used

- **CO₂ laser:** Most common; 10.6 μm wavelength; high power; for metals, plastics, wood
- **Nd:YAG laser:** 1.06 μm wavelength; pulsed; for metals and ceramics
- **Excimer laser:** UV range; for polymers and biological materials

LBM Advantages & Disadvantages

ADVANTAGES	DISADVANTAGES
Can machine ANY material — metals, ceramics, composites, polymers, glass	HAZ (Heat-Affected Zone) created
No tool wear; no physical contact	High equipment cost
High precision; narrow kerf width	Difficult to control depth accurately
Suitable for very thin and delicate materials	Molten metal re-solidification can affect quality
Fast processing speed for thin materials	Reflective materials (copper, aluminium) are difficult

LBM Applications

- Precision hole drilling (aircraft cooling holes, nozzle holes)
- Cutting sheet metal, ceramics, composites
- Laser welding and surface hardening
- Scribing and engraving
- Cutting and drilling of diamonds
- Medical: eye surgery (LASIK), tissue ablation
- Electronics: PCB patterning, semiconductor scribing

| ELECTRON BEAM MACHINING (EBM)

EBM uses a **focused beam of high-velocity electrons** to remove material by converting kinetic energy of electrons into thermal energy at the workpiece. Performed in **vacuum** to prevent electron scattering by air molecules.

Principle

- Electrons generated from electron gun (thermionic emission from heated tungsten cathode)

- Electrons accelerated by high voltage (25–200 kV)
- Beam focused by electromagnetic lenses to very small spot (<0.1 mm)
- Kinetic energy converts to heat at impact — temperature exceeds 6000°C
- Material melts and vaporizes (ablation)
- Must be done in vacuum (10^{−4} torr)

EBM Advantages & Disadvantages

ADVANTAGES	DISADVANTAGES
Extremely precise — very small holes possible (down to 0.025 mm)	Vacuum required — limits workpiece size and throughput
No tool — no tool wear	High equipment cost
Can machine any material (conductive or not)	HAZ present
Very high energy density — fast ablation	X-rays generated — radiation shielding required

EBM Applications

- Micro-hole drilling in aerospace components (turbine blade cooling holes)
- Welding in vacuum (used for reactive metals: titanium, zirconium)
- Lithography for semiconductor circuits
- Scribing brittle materials

I LBM VS EBM COMPARISON

FEATURE	LBM	EBM
Energy source	Photons (light)	Electrons
Medium	Atmospheric (no vacuum needed)	Vacuum required (10 ^{−4} torr)
Minimum hole size	~0.01 mm	~0.025 mm or smaller
Material restriction	All materials (reflective ones difficult)	All materials
X-ray generation	No	Yes — shielding required
Cost	High	Very high

 PREVIOUS YEAR QUESTIONS — UNIT 19 (LBM & EBM)

ETE NOV 2025 6 marks · CO5

QB3. Prepare a chart of various unconventional machining processes based on type of energy used (includes LBM and EBM).

See chart in Unit 17 PYQs above — includes LBM (Radiation/laser photons) and EBM (Radiation/high-velocity electrons).

 EXPECTED/SAMPLE QUESTIONS — UNIT 19

-  WARM 1. Describe the principle and process of Laser Beam Machining. (4 marks)
-  WARM 2. Why does EBM require vacuum? What happens if done in air? (2 marks)
-  HOT 3. Compare LBM and EBM on any five aspects. (5 marks)
-  LIKELY 4. What are the applications of LBM in aerospace and medical fields? (3 marks)
-  LIKELY 5. LASER stands for? What properties of laser make it suitable for machining? (3 marks)

UNIT 20 Additive Manufacturing (3D Printing)

I WHAT IS ADDITIVE MANUFACTURING?

Additive Manufacturing (AM) — also called **3D printing** — is a process of creating a three-dimensional object by **adding material layer by layer** from a digital CAD model. It is the opposite of subtractive manufacturing (machining).

 **Key concept:** AM builds parts by *adding* material layer-by-layer (additive). Machining *removes* material (subtractive). Casting and forming *shapes* material (formative).

I GENERAL AM PROCESS

1. **CAD Model:** 3D model created in CAD software (SolidWorks, AutoCAD)
2. **STL File:** CAD model converted to STL format (tessellated surface mesh)
3. **Slicing:** STL file sliced into thin layers (0.01–0.5 mm) by slicing software
4. **Build:** Machine builds part layer by layer
5. **Post-Processing:** Remove supports, clean, cure, or machine as needed

I ADDITIVE MANUFACTURING METHODS

1. Fused Deposition Modeling (FDM)

- **Material:** Thermoplastic filament (PLA, ABS, Nylon)
- **Process:** Filament heated and melted → extruded through nozzle → deposited on build platform → solidifies → next layer deposited on top
- **Advantages:** Low cost, easy to use, wide material range
- **Disadvantages:** Visible layer lines, anisotropic properties, requires support structures
- **Applications:** Prototypes, consumer products, jigs and fixtures

2. Stereolithography (SLA)

- **Material:** Photopolymer resin (UV-curable)
- **Process:** UV laser traces cross-section on resin surface → resin cures (polymerizes) → build platform moves down → next layer cured → part built upside-down from resin vat
- **Advantages:** Excellent surface finish, high accuracy, smooth parts
- **Disadvantages:** Limited materials, resins can be brittle, UV exposure needed after printing
- **Applications:** Dental models, jewellery, prototypes requiring high detail

3. Selective Laser Sintering (SLS)

- **Material:** Nylon, polyamide, or metal powder
- **Process:** Laser selectively sinters (fuses by heat) powder particles in each cross-section → build platform drops → new powder layer spread → process repeats
- **Advantages:** No support structures needed, strong functional parts, wide material range
- **Disadvantages:** Rough surface, slow, expensive equipment
- **Applications:** Functional prototypes, low-volume production parts, complex geometries

4. Selective Laser Melting (SLM) / Direct Metal Laser Sintering (DMLS)

- **Material:** Metal powders (stainless steel, titanium, aluminium, Inconel)
- **Process:** Similar to SLS but laser fully melts (not just sinters) metal powder
- **Advantages:** Full density metal parts, excellent mechanical properties, complex internal structures
- **Applications:** Aerospace components, medical implants, turbine blades

5. Binder Jetting

- Liquid binder selectively deposited on powder bed to bond particles
- Used for sand moulds, metal parts, full-colour models

6. Electron Beam Melting (EBM — AM version)

- Electron beam melts metal powder in vacuum
- Used for titanium implants (biomedical) and aerospace parts

I ADDITIVE MANUFACTURING VS CONVENTIONAL MACHINING

FEATURE	ADDITIVE MANUFACTURING	CONVENTIONAL MACHINING
Material approach	Adds material layer by layer (additive)	Removes material (subtractive)
Geometry capability	Any complex geometry including internal channels	Limited by tool access and fixturing
Material waste	Very little waste — near net shape	Significant material waste as chips
Setup time	Fast setup (digital file → print)	Requires fixtures, tooling, setups
Surface finish	Generally rough (layer-by-layer marks)	Excellent surface finish possible
Tolerances	Lower dimensional accuracy	Very high dimensional accuracy (microns)

FEATURE	ADDITIVE MANUFACTURING	CONVENTIONAL MACHINING
Production volume	Best for 1–100 units (prototyping)	Best for mass production
Tooling cost	No tooling required	Tooling and fixturing required
Material types	Plastics, metals, ceramics, composites, bio-materials	Primarily metals and plastics

I APPLICATIONS OF ADDITIVE MANUFACTURING

FIELD	APPLICATIONS
Aerospace	Turbine blades with internal cooling channels, structural components, fuel nozzles
Medical	Customized implants (hip, knee, dental), surgical guides, prosthetics, drug delivery
Automotive	Prototypes, low-volume parts, tooling jigs and fixtures
Defence	Rapid replacement parts, weapons components, drones
Construction	3D-printed buildings, bridges, concrete structures
Education	Visual models, teaching aids, research prototypes



PREVIOUS YEAR QUESTIONS — UNIT 20 (ADDITIVE MANUFACTURING)

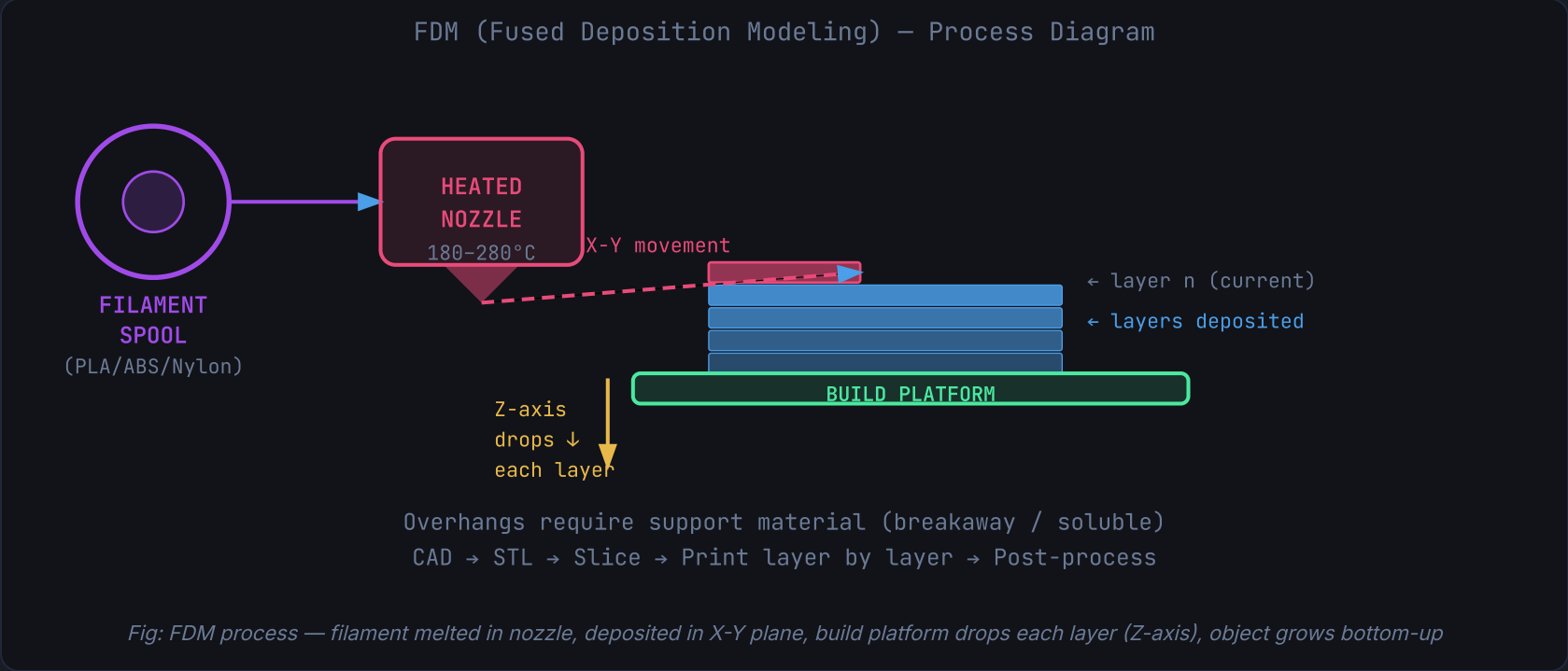
ETE NOV 2025

6 marks · CO5

QB5. Explain fused deposition modeling process of additive manufacturing.

FDM (Fused Deposition Modeling) Process:

- CAD to STL:** 3D CAD model converted to STL (stereolithography) file format — tessellates the surface.
- Slicing:** STL file sliced into thin horizontal layers (typically 0.1–0.3 mm) by slicing software.
- Setup:** Thermoplastic filament (PLA, ABS, Nylon) loaded into printer.
- Printing:** Filament fed into heated nozzle (180–280°C). Nozzle moves in X-Y plane depositing molten material in the pattern of each cross-section. Material solidifies quickly on contact with cooler built surface.
- Layer-by-layer:** Build platform moves down by one layer thickness (Z-axis). Next layer deposited on top of previous layer.
- Support Structures:** Overhanging features supported by breakaway or water-soluble support material.
- Post-Processing:** Remove supports, sand or polish if needed.



Applications: Rapid prototypes, consumer products, educational models, jigs, fixtures.

Advantages: Low cost, easy to use, large range of materials, can produce functional parts.

Disadvantages: Visible layer lines, anisotropic strength, requires support for overhangs.

RE-SESSIONAL NOV 2025

8 marks · CO5

Q8. Compare applications of additive manufacturing techniques in any two fields.

1. Medical / Healthcare:

- Customized implants (hip joint, knee joint, dental implants) made to exact patient anatomy from CT scans
- Surgical models and guides for planning complex surgeries
- Prosthetics and orthotics — lightweight, custom-fit
- Hearing aids — individually shaped for each patient
- Drug delivery devices and pharmaceutical dosage forms

2. Aerospace:

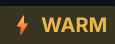
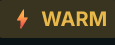
- Turbine blades with complex internal cooling channels — impossible to make by conventional machining
- Structural components with topology optimization — lightweight with same strength
- Fuel nozzles: GE Aviation's 3D-printed fuel nozzle has 25 parts consolidated into 1
- UAV/drone components — rapid replacement parts in field
- Rocket engine components using metal AM (SLM/DMLS)

RE-SESSIONAL APR 2026 3 marks · CO1

Q7. Differentiate between machining process and additive manufacturing process.

ASPECT	MACHINING	ADDITIVE MANUFACTURING
Nature	Subtractive — material removed	Additive — material added layer by layer
Material waste	High (chips discarded)	Very low (near net shape)
Complexity	Limited by tool access	Any geometry including internal features
Accuracy	Very high (µm range)	Lower (0.1–0.5 mm typical)
Production	Better for mass production	Better for prototypes and custom parts

💡 EXPECTED/SAMPLE QUESTIONS — UNIT 20

-  1. Describe Selective Laser Sintering (SLS) process. (4 marks)
-  2. What is Stereolithography (SLA)? What material does it use? (3 marks)
-  3. What is STL file? Why is it used in additive manufacturing? (2 marks)
-  4. What are the advantages of additive manufacturing over conventional manufacturing? (4 marks)
-  5. Why is additive manufacturing preferred in medical implant manufacturing? (3 marks)

⚡ QUICK REVISION — ALL ADVANCED MACHINING PROCESSES AT A GLANCE

PROCESS	ENERGY	MEDIUM	MATERIAL LIMITATION	BEST FOR
USM	Mechanical (20–40 kHz vibration)	Abrasive slurry (water + SiC/Al ₂ O ₃)	Hard, brittle, non-conductive	Glass, ceramics, quartz, carbides
AJM	Mechanical (high-velocity gas)	Gas + abrasive particles	Brittle materials (not soft/ductile)	Deburring, cleaning, thin brittle materials
ECM	Electrochemical (anodic dissolution)	Electrolyte (NaCl, NaNO ₃)	Must be electrically conductive	Complex shapes, no tool wear, superalloys
EDM	Thermal (spark erosion)	Dielectric (kerosene, DI water)	Must be electrically conductive	Dies, moulds, hardened steel
LBM	Radiation (laser photons)	Air/atmosphere	Any (reflective materials difficult)	Precision holes, cutting, scribing
EBM	Radiation (electron beam)	Vacuum required	Any material	Micro-holes, vacuum welding, semiconductors

